

Hex9: A Quasi-Authalic, Quasi-Continuous Hexagonal DGGS on the Reference Ellipsoid

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Abstract

Discrete global grid systems are conventionally designed over a prior coordinate reference system and inherit its compromises. Hex9 inverts the direction of design: we ask what requirements a hierarchical grid must satisfy to be geometrically coherent — intrinsic orientability, flat mode transport, vertex closure, refinement commutativity — and show that these requirements, taken together, essentially determine the grid. The admissible cell primitive is the triangle; the admissible seed is the octahedral triangulation of S^2 ; admissible refinement is uniquely aperture 9 — the only factor whose octant hexagonalises by half-hexagons, and uniquely up to chirality at that, so the hexagonal dual lattice admits exactly one orientation per chirality — both fixed by the same machine-verified exhaustive enumeration. The structure that survives is a shifted-aperture-9 hexagonal hierarchy in which every cell carries a unique canonical address derived from the construction alone: truncated at level L, the address is a DGGS zonal identifier in the sense of OGC Topic 21; carried to the limit, a function recovers from it a point on the reference ellipsoid to arbitrary precision. The addressing is quasi-continuous — position is recoverable everywhere except on a measure-zero set of seams — rather than continuous in the strict ISO 19111 sense; the same mathematical object serves as both zonal identifier and position-recovery coordinate, with no prior coordinate reference system as input. A separable geometric realisation — an analytical octahedral base projection composed with an optimal-transport-derived area-correcting warp — places the grid on WGS84 with quasi-uniform cell areas: at level 5, 99% of the 708,588 cells lie within 0.005% of ideal area, with residual deviation confined to the six octahedral vertices required by the topology. The combinatorial grid is projection- and ellipsoid-independent; only the warp is specific to the reference body, and it is recomputable for any ellipsoid, terrestrial or planetary.

Keywords: discrete global grid system (DGGS) · coordinate reference system (CRS) · hexagonal grid · octahedron · aperture 9 · optimal transport · equal-area projection · spatial indexing · WGS84

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54 Introduction

55 Discrete global grid systems face a set of familiar tradeoffs. Cell shape, aperture, orientation, and
56 global embedding each involve compromises between equal area, hierarchical consistency, polar
57 behaviour, and computational tractability — compromises often governed by a prior choice of
58 coordinate reference system, which in turn constrains what tradeoffs remain.

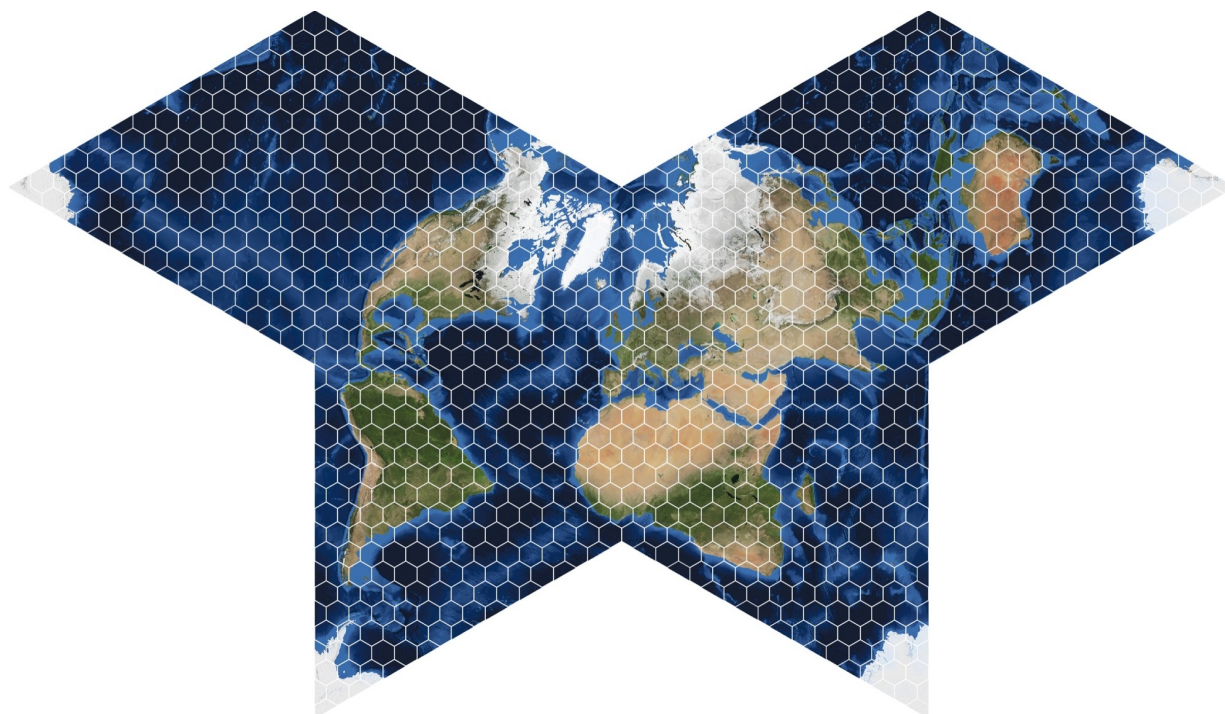


Figure 1: The Hex9 level-2 grid (972 cells) over NASA Blue Marble imagery, in the system’s native space: the `b_oct` octahedral “butterfly” net (§11). In `b_oct` every cell is a congruent regular hexagon (§11c) — the reshaping of the familiar continents is the work of the octahedral unfolding, not of the cells. Conventional reprojected views of the same grid appear in §14c.²

59 Hex9 is an exploration of a different resolution strategy: rather than choosing among tradeoffs,
60 we ask what requirements a grid must satisfy to be geometrically coherent — consistent at every
61 edge, every vertex, and every level of the hierarchy — and examine how far those requirements
62 determine the structure on their own.

63 These requirements significantly reduce the design freedom. A minimal set — orientability, parity
64 transport, vertex closure, and refinement commutativity — strongly constrains admissible cell
65 shape, embedding class, and aperture structure. What appears to be a large design space resolves
66 to a small family of solutions, selecting a restricted class up to global symmetry.

²Generated by `examples/ex0097_smp_grid.py` (butterfly:0500, L2).

This derivation has a direct consequence: because the structure is derived rather than chosen, every cell carries a distinct identity tied to a specific surface location. Cell identity serves as a self-contained locator — no prior coordinate reference system is needed to establish where a cell is.

The derivation was discovered rather than designed. The project began as a planar hexagonal tiling study and first targeted the icosahedron — the natural choice, given its near-spherical geometry and the precedent of existing icosahedral grids. The tiling could not be made globally consistent, and the failure resisted diagnosis: the obstruction is not geometric but topological. The odd valence of icosahedral vertices makes a consistent two-colouring of faces impossible, and that two-colouring turns out to be load-bearing for everything a coherent hexagonal hierarchy needs. Once the parity constraint was understood, the octahedron emerged by elimination as the unique Platonic solid with equilateral faces and even-valent vertices. The system presented here is the consequence of following that constraint to its conclusions.

The paper develops this argument constructively. The first part traces the coherence arc, showing step by step how each requirement eliminates alternatives. The second part describes the AK+Warp projection — Anders Kaseorg’s octahedral base projection composed with an area-correcting warp — the engineering path from the abstract octahedral structure to a quasi-authalic realisation on the reference ellipsoid (typically WGS84).

Axiom Set

Axiom 1 — Domain (Ellipsoidal Manifold)

The Earth is modelled as a smooth reference ellipsoid (e.g. WGS84), supporting a well-defined normal field and latitude/longitude parameterisation.

- This is the continuous substrate being discretised.
- No projection is privileged at this level.

Axiom 2 — Discrete Carrier (Simplicial Primacy)

The ellipsoid is discretised by a finite, recursively refinable 2D simplicial complex.

- The primitive cell is a triangle (2-simplex).
- All higher structure is derived from simplicial adjacency.
- No non-simplicial base cells exist in the canonical representation.

Consequence: hex-like structures are derived, not primitive.

Axiom 3 — Global Topological Regularity

The simplicial complex forms a valid triangulation of S^2 with:

- no boundary,
- no exceptional cell types,
- curvature expressed metrically, not topologically.

This enforces uniform representation class across the entire domain and excludes complexes with topological defects (such as forced pentagonal faces).

Consequence: combined with Axiom 2 and Axiom 4 — simplicial carrier, even valence from vertex closure, and this axiom’s uniformity requirement — the admissible complex is the octahedral triangulation.

Axiom 4 — Mode Transport

Assign to every face-to-face edge crossing in the triangulation the nontrivial element of $\mathbb{Z}_2$ — a binary flip. This transport is consistent when traversing any closed loop of crossings composes to the identity (trivial holonomy, in the language of differential geometry; equivalently, a flat $\mathbb{Z}_2$ connection, in gauge-theoretic terms; combinatorially, the face-adjacency graph is bipartite). Consistency is equivalent to the existence of a globally coherent binary face field: fix the value of any one face and every other face's value follows directly. We call this field the **mode** (developed in §§2–3).

Any refinement operator must preserve this consistency. Failure to do so introduces a transport defect.

Consequence (proved in §6): a refinement hexagonalises only if its octant tiles by half-hexagons, and aperture 9 (linear factor 3) is the unique factor that does so uniquely up to chirality — apertures 4, 16, 25, 49 cannot tile at all; 36, 81, ... tile in many inequivalent ways. Aperture 9 is therefore not merely minimal but the only refinement that determines a grid.

Axiom 5 — Refinement Invariance

Any refinement operator:

- preserves simplicial type,
- preserves adjacency relations,
- commutes with global indexing,
- preserves parity transport (Axiom 4).

Refinement never introduces new structural categories.

Axiom 6 — Canonical Orientation (Chirality Fixing)

The global orientation of simplices is fixed by a single consistent chirality choice induced by embedding into the reference ellipsoid frame.

- This resolves the residual $\mathbb{Z}_2$ symmetry left after Axioms 2–5.
- The choice is globally coherent, not locally independent.

Consequence: this is where the half-hexagon orientation freedom collapses to one surviving solution per chirality.

Axiom 7 — Geodetic Anchoring (Minimal Frame Fixing)

A minimal geodetic frame is fixed:

- poles define the primary axis,
- a single meridian defines longitudinal zero,
- together inducing a canonical partition of the domain.

This is not arbitrary once fixed — it is the reference gauge of the system.

Axiom 8 — Unique Cell Addressing

Every cell in the hierarchy has a unique canonical address, and every valid address identifies exactly one cell: the map between cells and canonical addresses is a bijection. This bijection is compatible with hierarchical structure: a cell's address encodes its position in the refinement hierarchy, and the parent–child relationship is recoverable directly from address structure. (Cells that straddle a parent boundary — the split hexagons of §10 — admit two parent lineages and hence two lineage descriptions; the canonical address is fixed by convention. Uniqueness

is a property of the addressing scheme, not of raw lineage paths, which §10 distinguishes from addresses for exactly this reason.)

Combinatorial position is read directly from the address: cell identity and the parent–child hierarchy require no transformation. Geographic location follows from that position through the system’s own canonical realisation (§11), a fixed map that adds no degrees of freedom — so no prior or external coordinate reference system is required as input. The grid fixes the discrete structure; the realisation supplies the metric embedding; the two are separable concerns (§11) composed to recover a location from an address.

Axiom 9 — Dual Consistency (Hex Emergence)

The dual graph of the simplicial complex induces a structured hexagonal lattice:

- hexagons are derived dual cells,
- the total topological defect of 12 required by the Euler characteristic of S^2 (Σ over vertices of $(6 - \text{valence}) = 12$) is concentrated at the 6 seed vertices — a deficit of 2 at each — rather than appearing as face anomalies,
- no independent hex tiling is assumed at the primitive level.

Hex9 structure is emergent from simplices, not fundamental.

Main Theorem

Axioms 1–9 admit a discrete global grid that is **unique up to two free choices** — a global chirality (Axiom 6) and a geodetic gauge (Axiom 7). That grid is **Hex9**: the shifted-aperture-9 hexagonal grid on the octahedral triangulation of the reference ellipsoid, with the cell hierarchy and addressing developed in §§6–10.

Concretely, the axioms force each link of the following chain, and the arc sections that follow constitute its proof:

1. **Carrier.** The discrete carrier is a simplicial complex, and the only triangulation of S^2 with even, uniform valence and no exceptional cells is the octahedral triangulation (Axioms 2–4; §§1–5).
2. **Aperture.** A hexagonalisation requires the octant to tile by half-hexagons, and **only aperture 9 (linear factor 3) does so uniquely up to chirality** — the same exhaustive enumeration that fixes the chiral pair (Axiom 6). Apertures 4, 16, 25, 49 cannot tile at all; 36, 81, ... tile in many inequivalent ways. Aperture 9 is the unique refinement that determines a grid (Axioms 4–6; §6).
3. **Hex emergence.** The dual carries the 12 units of topological defect required by the Euler characteristic of S^2 as twelve five-neighbour hexagons at the six octahedral vertices, not as a distinct cell type (Axiom 9; §7).
4. **Orientation.** The residual orientation freedom collapses to a single chiral pair (Axiom 6; §8).
5. **Identity.** Every cell carries a unique canonical address from which its geographic location is read directly, with no prior coordinate reference system (Axiom 8; §§9–10).

The geometric realisation on the ellipsoid (§11) is a separable engineering step: it places this combinatorial object onto WGS84 but adds no degrees of freedom to the structure the axioms determine.

1. The Simplicial Carrier

A discrete global grid partitions the globe into a finite collection of cells. The choice of cell shape is the first apparent design decision: squares, triangles, hexagons, and other polygons all tile the plane, and several tile the globe with appropriate modifications.

The triangle stands apart from this class. It is the minimal polygon — three edges, three vertices, no simpler closed shape exists — and the only one whose orientation datum is purely combinatorial: an ordering of its three vertices determines an orientation, two orderings being equivalent when they differ by an even permutation, with no metric or ambient structure required. (Which orientation class reads as *clockwise* is decided by the orientation of the surface the triangle sits in — the labels need a viewing side — but the datum itself lives in the vertex ordering alone.) Every other polygon either decomposes into triangles or carries orientation only through additional structure: a cyclic boundary order, a metric, or an embedding.

This intrinsic orientability makes the triangle the natural substrate for a globally coherent grid. We take the triangular field — a tiling of the globe by triangles — as the simplicial carrier on which the remaining coherence requirements will act.

A coherent grid carrier must satisfy several requirements simultaneously: operators such as interpolation and refinement must be definable without auxiliary choices; each cell must support a unique, intrinsic coordinate system; refinement must be closed and structure-preserving; and no cell-type exceptions or singular vertices may appear. The triangle is the only 2-cell that satisfies all of these at once. Its barycentric coordinates are intrinsic — defined by the vertices alone, with no ambient metric required. It is closed under subdivision. It forms a simplicial complex without diagonal ambiguity.

On curved surfaces this distinction becomes decisive. A triangulation absorbs curvature at its vertices without changing cell type; tilings by higher polygons must place their topological obligations somewhere visible — as exceptional cells (the twelve pentagons of icosahedral hexagonal grids) or exceptional corners (the eight 3-valent corners of cube-based quadrilateral grids). The triangular field is therefore not one choice among many — every other option either reduces to it or requires additional structure not established by the coherence requirements alone.

2. Mode

In any triangulation of a surface without boundary, every edge is shared by exactly two triangles. In the familiar planar lattice those two triangles are always one *up* and one *down*: the faces fall into two classes, and every edge joins one face of each class. We call such a two-colouring of faces — adjacent faces always of opposite class — the **mode** of the triangular field: mode 0 for the down (negative) class, mode 1 for the up (positive) class.

Two cautions keep this definition honest. First, mode is not orientation. A coherent orientation — every face wound so that a shared edge is traversed in opposite directions by its two faces — exists on any orientable surface, and under it adjacent faces have the *same* handedness: in the planar lattice, up and down triangles are both counterclockwise. Mode is an additional structure, a bipartition of the face-adjacency graph. Second, its existence is not automatic. The boundary of the 3-simplex — four triangles, orientable, coherently orientable — admits no mode field: its face-adjacency graph is K_4 , which has no proper two-colouring. The icosahedron, with five faces at every vertex, fails the same way. Mode exists exactly when the face-adjacency graph is bipartite; which triangulations achieve this is the substance of §§3–4.

Where a mode field exists it is essentially unique. Fix the mode of any one face, and the mode of every other face follows directly; the only freedom is which class is named 0, a global reflection.

Every face carries a mode value; every interior edge is incident to exactly two faces of opposite mode; the field is bipartite on its faces.

Mode is therefore the first global coherence requirement of the system, not a free consequence of the carrier — which is precisely why Axiom 4 must demand it. Every grid built over a triangular field either possesses this invariant and preserves it across refinement and adjacency, or lacks it and pays with defects. Every edge crossing is a mode-flipping step; what follows from global consistency of those steps is the subject of §3.

3. Mode Transport

Section 2 defined mode where it exists and showed that its existence is a genuine constraint. To pose the question uniformly — on any triangulation, whether or not a mode field exists — we work with the transport rather than the field: the transport operator on each edge is the mode flip, the demand that the two faces sharing the edge take opposite classes. It is combinatorial in origin: no metric or embedding is required to define it.

We now ask what it means for this transport to be globally consistent.

Define a **mode transport** as a rule that assigns, to any path through the triangular field (a sequence of edge crossings from face to face), a net mode change in $\mathbb{Z}_2$. The transport is **consistent** if the net change depends only on the endpoints of the path — not on the route taken; equivalently, traversing any closed loop returns the mode to its starting value. This is the condition differential geometry calls trivial holonomy and gauge theory calls a flat $\mathbb{Z}_2$ connection; we will simply say the transport is **flat**. On any connected triangulation flatness is equivalent to the existence of a globally consistent mode field: fix the mode of any one face and every other face's mode follows. Any refinement must preserve flatness; otherwise it introduces a transport defect.

The requirement becomes non-trivial when we demand that consistency extends to all refinements of the triangulation. A refinement introduces new faces, edges, and vertices — and the transport must remain consistent at every scale. This is the content of Axiom 4, and its consequences for the admissible aperture class are developed in §6.

What the transport implies immediately is simpler: any structure built over the triangular field — any labelling, orientation, or subdivision — must account for the mode flip at every edge crossing, or introduce a defect. The mode is not merely a local colouring; it is a global invariant that any grid must either propagate correctly or break.

At individual edges the transport is binary and clean. At vertices — where multiple edges meet — the accumulated transports must also close consistently. What this implies for the global topology is the subject of §4.

4. Vertex Closure

Before the argument, its conclusion: we do not pick the octahedron from a menu of base shapes. It is the only one on which a true hexagonal grid can both wrap a closed surface and unfold flat to the plane without contradiction. What can read as a self-imposed constraint is really the absence of alternatives — the rounder shapes other grids build on do not support the same coherent hexagonal hierarchy.

Section 3 established that mode transport must be flat — any closed loop must return the mode to its starting value. Vertices are where this condition is most constraining: at every vertex, a ring of triangular faces meets, and traversing that ring forms a closed loop.

Consider a vertex where k triangular faces meet (valence k). Moving from face to face around the vertex traverses exactly k edges. Each edge crossing carries a mode flip. For the transport around this closed loop to return to identity, the total number of flips must be even — requiring k to be even.

This is the **vertex closure condition**: it selects triangulations in which every vertex has even valence. It ensures that the face adjacency graph is bipartite — no odd cycles in the dual.

Vertex closure alone does not finish the argument. It requires every valence to be even, but not that all valences be equal: triangulations of S^2 that mix different even valences also carry a consistent mode field. A concrete example is the midpoint-subdivided octahedron — original vertices of valence 4, edge-midpoint vertices of valence 6, every valence even, dual bipartite, mode well defined.³ We make no claim that such mixed-valence triangulations fail under refinement; flatness by itself admits them.

What removes them is the regularity requirement of Axiom 3, not a theorem. The star of a vertex is its ring of incident faces, and in a triangulation the star’s combinatorial type is fixed by the valence alone: two vertex stars are of the same type exactly when the same number of triangles meets at each. Axiom 3’s “no distinguished vertex classes” is precisely the demand that all vertex stars be of one type. A triangulation mixing valences 4 and 6 has two, and every structure built over it — refinement rules, addressing, adjacency — would have to case-split on them; with a single star type, every vertex behaves identically under refinement and addressing. Uniform even valence is therefore imposed as a regularity condition, not derived as a topological necessity.

Given uniform even valence, the Euler characteristic of S^2 constrains what is possible. For a triangulation with V vertices, E edges, F faces:

$$V - E + F = 2, \quad \text{with} \quad E = 3F/2$$

For uniform valence v the relation $2E = vV$ gives:

$$V = 12 / (6 - v)$$

For $v = 4$ (the minimum even valence greater than 2): $V = 6$, $F = 8$, $E = 12$. For $v = 6$: the denominator vanishes — uniform valence 6 cannot close on the globe. For $v \geq 8$: the formula yields a negative vertex count — impossible.

The octahedral triangulation — $V=6$, $E=12$, $F=8$, uniform valence 4 — is the one abstract triangulation of S^2 satisfying all conditions. The division of labour is explicit: vertex closure, a derived condition, forces even valence; regularity (Axiom 3), a stated requirement, forces uniform valence; the Euler characteristic then leaves $v = 4$ as the only possibility.

Consequence: a $\mathbb{Z}_2$ face-mode can be defined on any triangulation of S^2 with a bipartite dual graph — equivalently, on any triangulation in which every vertex has even valence. Vertex closure is the derived part of this argument; uniformity of valence is required by Axiom 3 rather than derived. Together they admit exactly one triangulation of S^2 : the octahedral one.

5. The Octahedral Embedding

Section 4 established that the minimal regular seed complex compatible with mode transport on the globe has exactly 6 vertices, 12 edges, and 8 triangular faces, with uniform vertex valence 4. There is exactly one abstract triangulation of S^2 with this combinatorial structure — and it

³Combinatorially this example is the aperture-4 (midpoint) refinement of the octahedral seed — the same aperture §6 rejects. There is no tension: as a triangulation it preserves flat mode transport, since every valence stays even. What it fails is the separate hexagonalisation requirement developed in §6 — its octant does not tile by half-hexagons.

is the octahedral triangulation. Its geometric realisation as a convex polyhedron is the regular octahedron.

On the sphere, every embedding of the octahedral seed is equivalent: the sphere’s symmetry group $O(3)$ contains the full octahedral group O_h in any orientation, and no embedding is preferred. The reference ellipsoid breaks this, and the more the body’s symmetry breaks, the more of the embedding it pins. Geodetic reference ellipsoids (WGS84, GRS80) are ellipsoids of revolution: such a body retains only the symmetries fixing its polar axis — continuous rotation about the axis, the equatorial mirror, and the vertical mirror planes (the group $D_{\infty h}$). The poles are the fixed points of this residual symmetry.

An embedded octahedron shares with the ellipsoid exactly those symmetries common to both, and the shared group depends on which octahedral axis is aligned with the polar axis. Aligning a vertex pair (a 4-fold axis) retains D_{4h} , of order 16; a face pair (a 3-fold axis) retains D_{3d} , of order 12; an edge pair retains D_{2h} , of order 8; a generic orientation retains almost nothing. Pole-on-vertex anchoring is therefore not one choice among equals: it is the unique orientation class preserving the maximal common symmetry of seed and surface.⁴

This maximal residual symmetry is what the construction uses, and it does two distinct jobs. Combinatorially, it anchors the seed: under D_{4h} the 8 octant faces form a single orbit — every octant face is equivalent to every other — so a single projection function serves all 8 octants, with mode-1 faces obtained from mode-0 by one reflection ($y \rightarrow -y$). The same orbit then carries into the realisation (§11): the area-distortion field the warp must correct shares the body’s symmetry, so one warp field, computed once, serves every octant, and the correction is exactly uniform across octants — no octant pays more warp than another, a guarantee an asymmetric embedding could not make. The anchoring also reduces the continuous rotational gauge freedom to the discrete 4-fold rotation, which the meridian anchoring of Axiom 7 resolves. Pole anchoring does not impose a coordinate system; it selects the embedding in which the seed inherits the most structure from the surface.

The result is a canonical partition of the globe into 8 triangular octants — not a projection choice, but the direct consequence of mode transport closure applied to a closed orientable surface. The 8 octants are the top-level cells of a grid hierarchy following these constraints. Further structure — refinement, orientation, dual cells — is built over this seed. The octahedral embedding is not one possible global frame among many; it is the frame the coherence requirements construct.

Is this a loss of flexibility? It can read as one — why commit to a single base polyhedron when others give rounder cells? — but the flexibility is largely illusory. No hexagonal grid tiles a sphere at all without defects (Euler), and demanding one that *both* closes on a folded polyhedron *and* unfolds coherently to the plane, carrying the same two-colouring through every refinement, is a genuinely narrow requirement: §§4–5 show the octahedron is the seed that survives, not one rounder option among many that we set aside. What the commitment then buys is the coherence the rest of the paper rests on — the two-colouring (mode transport) is what produces the half-hexagon hierarchy and its exact, self-inverting nesting (§7, §10e, §13b), the property §12 sets against iterated hexagonal indices — together with an economy of realisation: the eight octant faces are all equivalent by symmetry, so a single projection and warp serve the whole globe (§11), and the unavoidable defect sits at six vertices rather than the twelve of an icosahedral grid.

⁴On a triaxial ellipsoid the residual symmetry drops to the finite group D_{2h} (order 8) and all three principal axes become salient. The maximal-common-symmetry principle still applies and pins the embedding up to a finite choice — aligning the body’s three 2-fold axes with 2-fold axes of the octahedron shares the full D_{2h} — but it no longer selects uniquely: the vertex-aligned embedding and the one rotated 45° about the polar axis (equatorial *edge* axes onto the equatorial principal axes) share the same group, so a convention is required — most naturally the continuous limit of the oblate case, which is vertex alignment. The eight octants remain a single D_{2h} orbit, so the one-projection, one-warp economy survives, and the realisation of §11 computes its warp numerically for whatever body is supplied.

The one real cost is cell shape — an octahedral seed is less round per cell than icosahedral or dodecahedral grids (aspect 1.37 vs H3’s 1.06; §12) — but those rounder grids do not deliver the same coherent, exactly-nested hexagonal hierarchy. The choice is between different objects, not a menu of equivalents.

6. Refinement Commutativity

The octahedral seed established in §5 defines a base simplicial complex on which a hierarchical refinement operator acts. Each triangular face is subdivided into k^2 child triangles by linear refinement at scale factor k . The refinement operator must preserve simplicial type, adjacency relations, and the flatness of mode transport (§3), and must commute with the global indexing of the mesh.

Refinement is chosen to commute with mode transport because mode is the global coherence field of the system. Without commutativity, refinement would not preserve identity under scale, and hierarchical addressing would cease to be stable under composition. This choice is not geometrically mandatory, but it is required for refinement to function as a consistent coordinate extension rather than a sequence of unrelated discretisations.

These requirements constrain admissible values of k . First, k must be a positive integer to ensure that refinement is a well-defined subdivision of the simplicial structure into congruent refinement classes.

Second, refinements that preserve mode transport consistency are those in which the mode of every child triangle agrees with the mode inherited from its parent at every induced edge (a homomorphism of the $\mathbb{Z}_2$ transport structure, in algebraic terms).

The mode two-colouring alone does not single out a factor: a parent’s children carry both modes at every scale, for every k , so mode transport is consistently inherited regardless. What singles out the aperture is a stronger requirement — that the refinement actually **hexagonalise**, and do so unambiguously.

Two conditions settle it, and a single exhaustive enumeration supplies both (verified by `experimental/halfhex_verify.py`, swept over k by `experimental/aperture_tiling.py`). First, the subdivided octant — a side- k triangle of k^2 triangular cells — must tile by half-hexagons (the `d_cells` of §7), each three cells, which is possible only when 3 divides k . Second, and decisively, that tiling must be **unique up to chirality**; otherwise the refinement admits several inequivalent grids and addressing is not stable under composition.

The enumeration is sharp. Apertures 4, 16, 25 and 49 (factors 2, 4, 5, 7) cannot tile the octant by half-hexagons at all. Apertures 36, 81 and larger *can* tile, but in many inequivalent ways — 220 distinct tilings already at aperture 36 — fixing no canonical grid. **Aperture 9 alone (factor 3) tiles uniquely up to chirality: exactly two tilings, a single mirror pair.** This is not a separate fact from the chirality result — it is the same enumeration: the two tilings *are* the chiral pair that Axiom 6 resolves (revisited in §8).

Aperture 9 is therefore not merely the smallest admissible refinement; it is the only one that determines a grid. We take $k = 3$ as the base refinement operator, and compose it for finer scales, yielding the aperture-9 hierarchy.

In the $k=3$ case, refinement introduces a systematic lateral displacement of child simplex centroids relative to the parent geometry. This displacement is not an artefact of embedding, but the geometric manifestation of parity-preserving refinement under $\mathbb{Z}_2$ transport. In the dual structure, this offset becomes visible as the characteristic shift in the induced hexagonal lattice, giving rise to the **shifted-aperture-9 hierarchy**. Intuitively, the parent’s centroid falls on the shared

long edge of its two half-hexagons rather than on any one child, so the nine children cannot sit concentrically; they take the offset arrangement of §8, and that offset is the displacement.

7. Dual Projection — Hexagonal Structure

The dual of a triangulation exchanges faces and vertices: each triangular face becomes a dual node, and each vertex becomes a dual face whose valence equals the vertex degree.

In the refined octahedral triangulation, interior vertices attain valence 6, producing regular hexagonal dual cells. The six vertices of the octahedral seed constitute the only non-uniform elements of the triangulation; each is 4-valent, a vertex deficit of 2, together accounting for the total defect of 12 required by the Euler characteristic (§4). The dual cells remain hexagons throughout: the deficit is carried not by a different cell shape but by twelve hexagons — two meeting at each seed vertex — that there have only five distinct neighbours, two of their sides folding together across the vertex. Refinement spreads regular valence-6 structure throughout the interior while localising this irregularity to the six seed vertices.

It follows that the hexagonal lattice is not imposed but emerges as the dual of the locally regular simplicial field. The octahedral embedding provides the minimal finite source of irregularity required by the Euler characteristic of S^2 , while preserving maximal hexagonal regularity elsewhere.

Within each octant, the dual lattice forms a half-hexagonal region (the **half-hexagon**) bounded by seed edges. The $k=3$ refinement introduces a systematic lateral displacement of these regions in the dual lattice. This shift is the spatial manifestation of enforcing refinement as a homomorphism of the mode transport structure.

The planar structure has a crystallographic signature. The hexagonal tiling cut along the half-hexagon long edges in the Hex9 arrangement has wallpaper group $p31m$ — a strict reduction from the $p6m$ of the uncut tiling. Colouring by mode reduces it further: every mirror of $p31m$ exchanges the two modes (a mode-preserving mirror would make the tiling achiral, contradicting the chiral pair of §8), so the mode-preserving subgroup is exactly the rotational part, $p3$. The $\mathbb{Z}_2$ of §2 is, in crystallographic terms, the quotient $p31m / p3$.

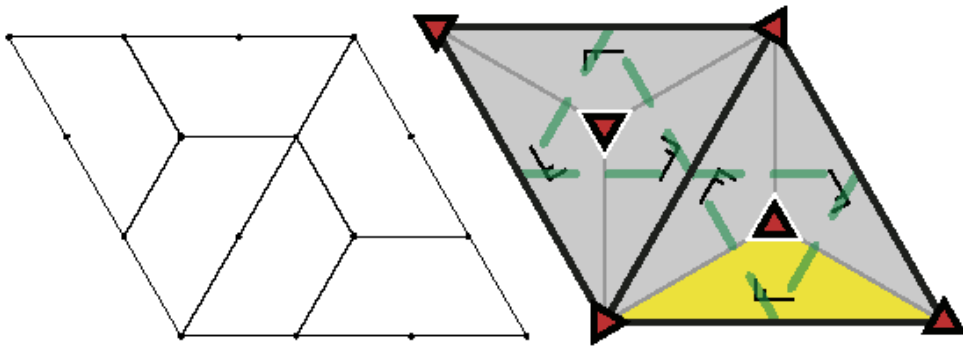


Figure 2: Crystallographic structure of the d_cell tiling: the translational unit (left) and its IUC symmetry — three-fold centres, mirror and glide lines, fundamental domain (right). The symmetry-reduction chain is $p6m \rightarrow p31m \rightarrow p3$ (§7).

The resulting system consists of a hexagonal dual lattice with a minimal, seed-localised defect structure and a refinement-induced shift. The remaining degree of freedom is the global and inter-octant orientation of this lattice.

8. Orientation Selection

The half-hexagon established in §7 is the natural octant carrier — the region of the dual hexagonal lattice bounded by the three seed edges of a single octant face. Its internal structure admits a further decomposition: the half-hexagon divides into three equilateral sub-regions, each corresponding to one third of the octant face.

Each sub-region can be independently oriented in two ways: the hexagonal cells within it may be arranged in one of two configurations related by reflection. With three sub-regions and two choices each, there are $2^3 = 8$ candidate orientations for the half-hexagon as a whole.

The mode transport constraint — specifically, the vertex closure condition established in §4 — acts on the internal boundaries between sub-regions. At every vertex where two or more sub-regions meet, the mode transport around that vertex must return to identity: trivial holonomy. This is the same condition that selects for even valence globally; applied to the internal half-hexagon boundaries, it filters the 8 candidates.

Exactly 2 of the 8 combinations satisfy the internal closure condition at every boundary vertex. The two survivors are related by a global reflection — a chiral pair, geometrically distinct but structurally equivalent up to handedness.

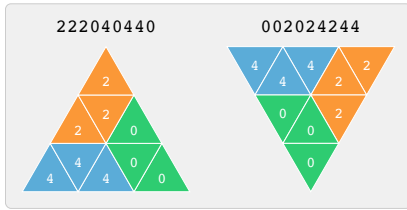


Figure 3: The surviving chiral pair: the two orientations of the 9-cell equilateral that satisfy internal closure (three half-hexagons each, coloured by orientation class). The members are mirror images — the residual chirality Axiom 6 fixes.⁶

This count is not an assertion. The full enumeration is machine-verified by `experimental/halfhex_verify.py` (checks V0–V6): of the 49 distinct hextile solutions (24 chiral pairs + 1 self-mirror), the long-edge constraint (A) admits 18, the three-equilateral structural constraint (B) admits 8, and their intersection $A \cap B$ is exactly the recorded Hex9 chiral pair. The constructive $2^3 \rightarrow 2$ argument above and this enumeration agree.

Axiom 6 resolves the remaining freedom: a single consistent chirality choice, induced by the embedding into the reference ellipsoid frame, selects one of the two. The orientation of the half-hexagon — and with it the entire hexagonal lattice across all 8 octants — is determined: no further choice remains.

This closes the enumeration. Each cell in the hierarchy is fixed by its octant, its refinement level, and its index within that octant at that level. No two cells share the same address; no address refers to more than one cell. The hexagonal lattice is globally indexed with no residual ambiguity. What follows from this is the subject of §9.

⁶Generated by `experimental/halfhex_further.py`; enumeration counts verified by `experimental/halfhex_verify.py`.

⁸Generated by `experimental/halfhex_further.py`; counts machine-verified by `experimental/halfhex_verify.py`.

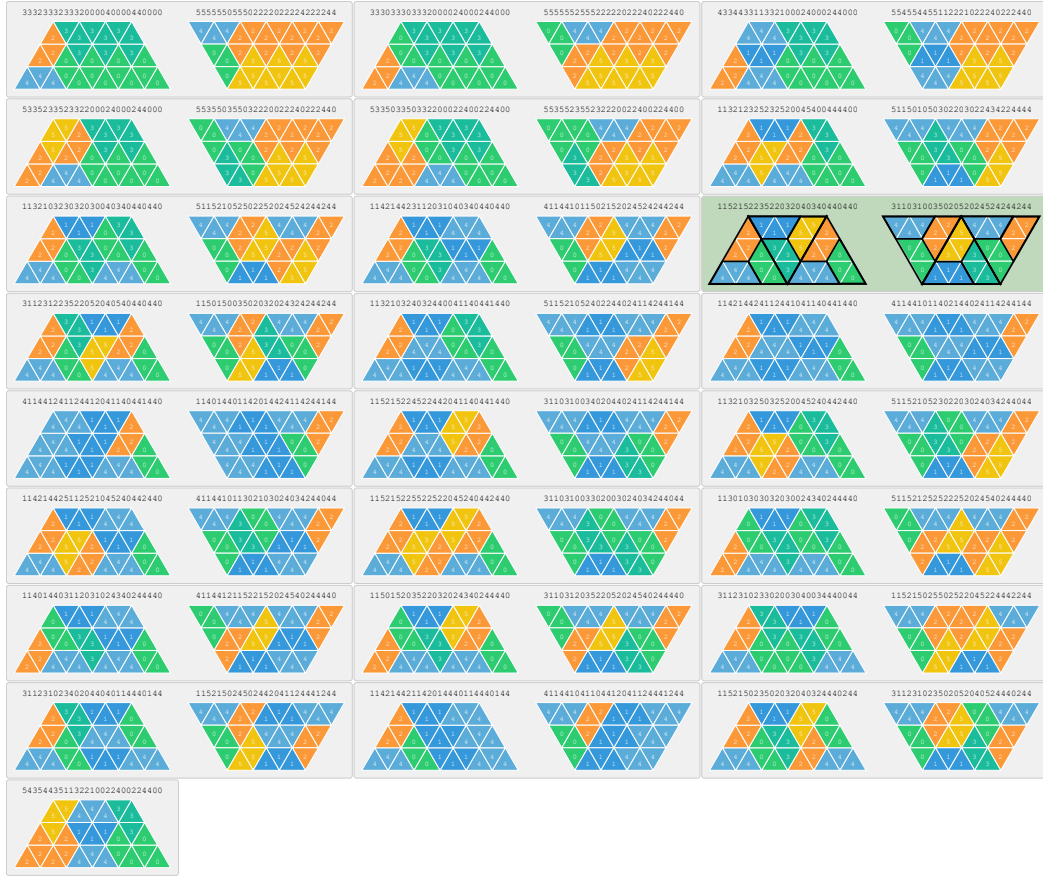


Figure 4: The 49 distinct hextile solutions, chiral pairs grouped and the self-mirror marked, with the Hex9 pair highlighted. Of $49 = 24 \text{ pairs} + 1 \text{ self-mirror}$, constraint A admits 18, constraint B admits 8, and $A \cap B$ is the highlighted pair.⁸

9. Hex9 Cell Identity — Structure as Locator

The system defined by Axioms 1–9 and constructed through the preceding steps is **Hex9**: a shifted-aperture-9 hexagonal grid on an octahedral embedding of the reference ellipsoid, with a cell hierarchy in which every cell has a distinct address, derived from simplicial coherence requirements alone.

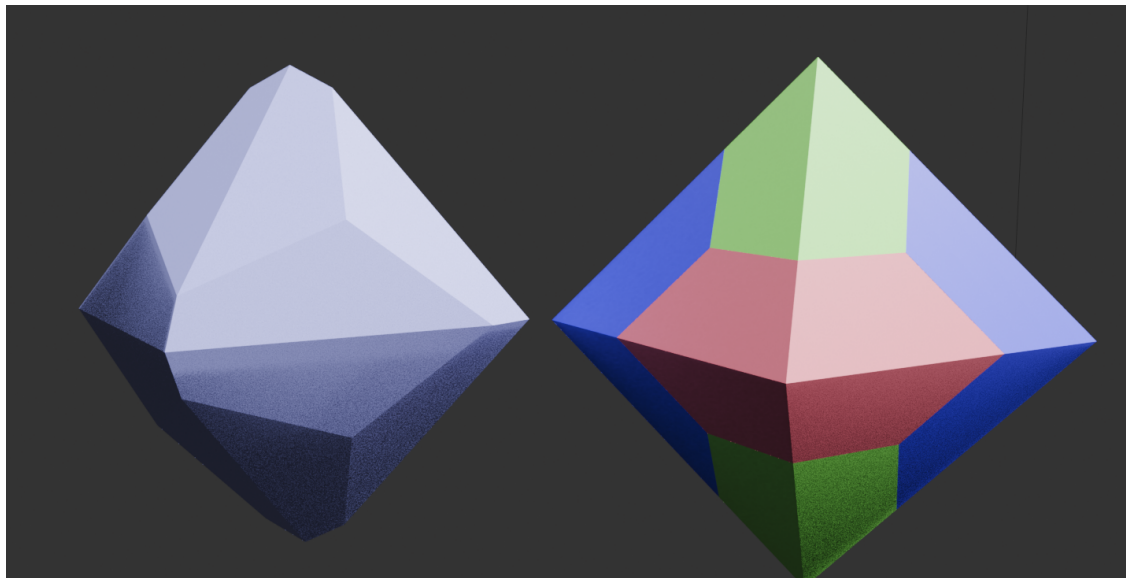


Figure 5: The seed solid and the 12 root cells. Left: the octahedron with each octant face creased into its three `d_cell` facets (24 faces, the diploid form that names the `d_cell`). Right: coloured per root `x_cell`, hue by octahedral axis, light/dark for the mode-0/mode-1 halves — at L0 every root cell is one of the 12 five-neighbour hexagons that meet at the octahedral vertices (appearing pentagonal once projected onto a reference ellipsoid). The faceting is illustrative; the cells live on the smooth ellipsoid.

Each Hex9 cell is identified by its path through the refinement hierarchy. This path is not a coordinate computed from a prior reference system — it is the cell’s identity. The bijection of Axiom 8 (Unique Cell Addressing) guarantees that this identity encodes a specific, unambiguous geographic region. At each finite refinement level the correspondence between addresses and cells is exact: one address, one cell, no exceptions. Cell location is recovered directly from combinatorial structure; no external coordinate transformation is required to establish it.

Hex9 is a discrete spatial reference system in the sense of OGC Abstract Specification Topic 21 ([Open Geospatial Consortium 2021](#)), which defines a DGGS as an integrated system comprising a hierarchy of discrete global grids, spatio-temporal referencing by zonal identifiers, and functions for quantization and zonal query (its clause 4.13; Appendix B maps the vocabularies). What distinguishes Hex9 within this class is the direction of derivation. The conventional workflow — select a coordinate reference system, then design a grid over it — is inverted. Here, the geometric coherence requirements of §§1–8 define the structure; the structure constitutes the locating system. No prior CRS is required as input.

This correspondence is discrete, not continuous. Hex9 cell identity uniquely identifies a geographic region at each resolution. The hierarchy of nested regions converges to a point as refinement increases, but Hex9 does not provide continuous coordinates in the sense of a projected CRS. For metric operations — distances, areas, interpolation — a geometric realisation onto the reference ellipsoid remains necessary, and is the subject of §11.

The enumeration completed in §8 is what makes self-contained location possible. A grid with

residual design freedom — orientations, apertures, or embeddings left as choices — cannot serve as its own locating system, because different choices produce different grids that disagree on cell identity. Hex9 has no residual freedom. Every cell is where it is because every constraint converges on that location.

10. Addressing and Continuity

10.0 Notation — the c/t/d/x grid taxonomy

The constructions of §§6–8 generate four overlapping grid spaces, designated by single letters: **c**, **t**, **d**, **x**. The sections that follow use this vocabulary constantly; this section fixes it. (The authoritative glossary, maintained alongside the implementation, extends these definitions.)

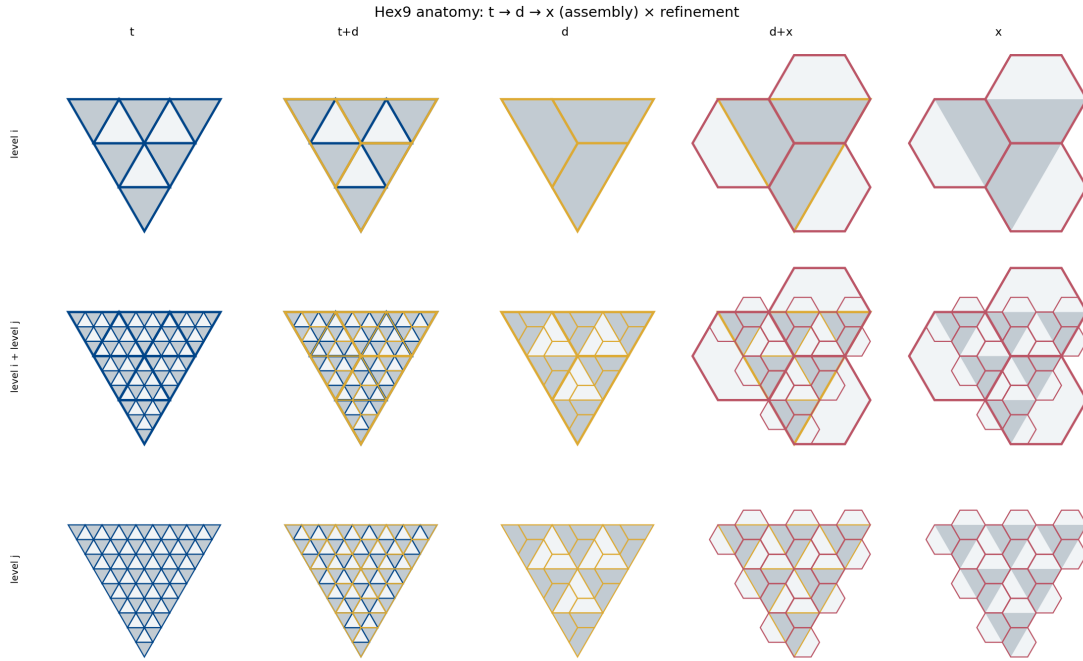


Figure 6: Anatomy of one parent triangle: **t**_cells, **c2** edges, **d**_cells, and the assembled **x**_cells with **x**_dig labels — the c/t/d/x taxonomy in one picture.¹⁰

t_cell — a triangular cell of the refined field: the working unit of the simplicial carrier. Each **t**_cell carries an intrinsic mode ($0 = \nabla$, $1 = \Lambda$; §2) and subdivides into 9 child **t**_cells at the next level (§6).

c2 — the label (0, 1, 2) of a **t**_cell edge, assigned by edge gradient: 0 = flat (horizontal), 1 = forward (positive slope), 2 = back (negative slope). Adjacent triangles agree on the **c2** value of their common edge, and the **c2** progression around any triangle is clockwise regardless of mode.¹¹

c_cell — a slot in the 96-position classifier grid: the raw output of the three-family inequality classification of §10a. Twelve of the 96 slots are the in-scope **t**_cell classes; the remainder are out of scope by construction.

¹⁰Generated by `examples/ex0400_anatomy.py`.

¹¹*Etymology*: “c2” is shorthand for “colouring 2” — one of the two distinct three-colourings of the wallpaper group p31m (Grünbaum & Shephard 1987 (Grünbaum and Shephard 1987), *Tilings and Patterns*, §8.3 “Color Pattern Types”, p. 433, type IH38; PP25[3]₁/p31m[3]₁ and PP25[3]₂/p31m[3]₂ — reproduced in the figure, the second with the Hex9 **d**_cell overlay aligned). A very early naming mismatch: despite the “2”, the Hex9 **c2** labelling actually corresponds to Grünbaum & Shephard’s *first* colouring, p31m[3]₁, not the second.

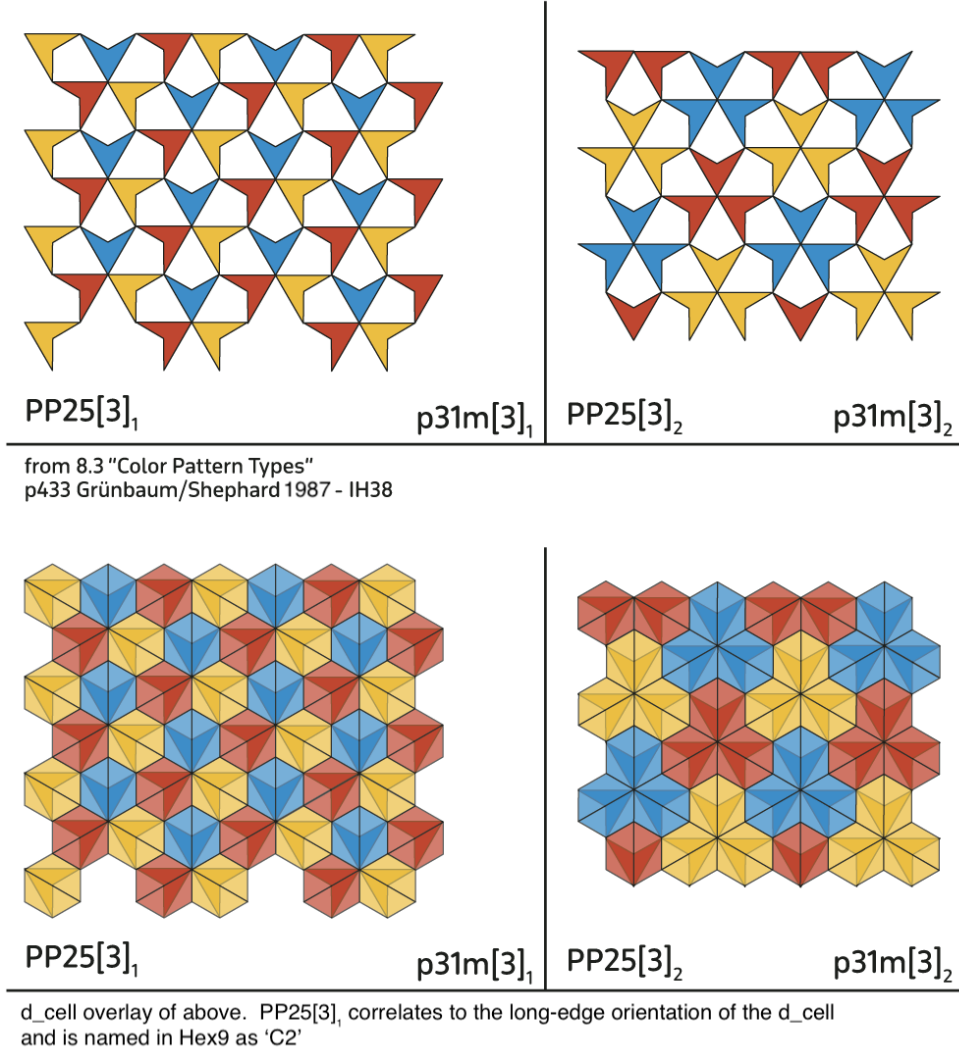


Figure 7: The two three-colourings of the p31m arrowhead tiling, PP25[3]₁ and PP25[3]₂ (top pair, redrawn after Grünbaum & Shephard ([Grünbaum and Shephard 1987](#)), §8.3 "Color Pattern Types", p. 433, type IH38), and the same colourings with the Hex9 d_{cell} tiling overlaid and aligned (bottom pair). Despite its name, the Hex9 c2 labelling corresponds to the *first* colouring, p31m[3]₁.

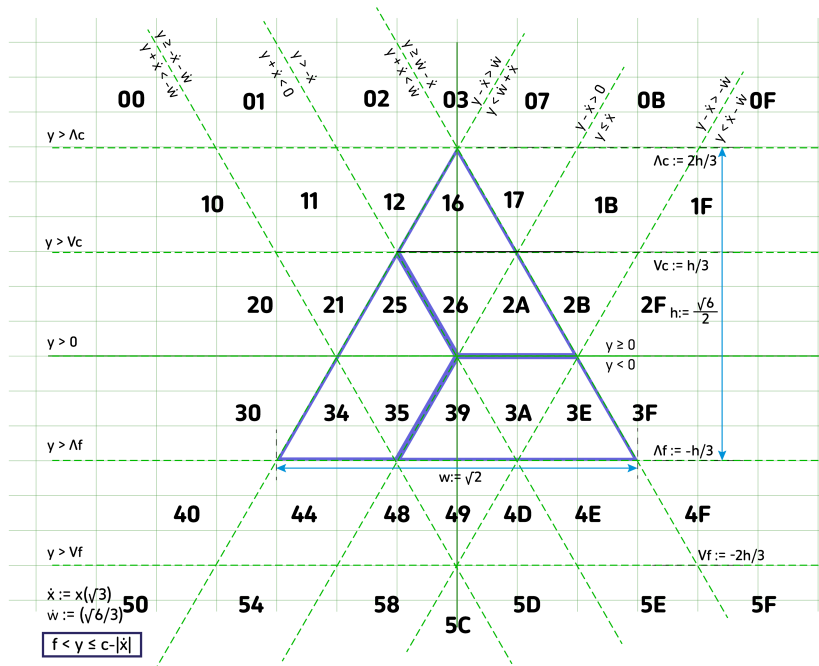


Figure 8: The classifier (c-layer): three band families partition the octant plane; their indices compose the 96-slot c _grid digit, which maps to a t _cell and thence to d _cell and x _cell. Locating a point is band membership, not search (§10a).¹³

515 **d_cell** (half-hexagon) — three t_cells grouped by their shared long edge. The c2 value of that
 516 long edge is the d_cell's digit ($d_dig \in \{0, 1, 2\}$), and the d_cell inherits the mode of its t_cells.
 517 The d_cell is the fundamental domain of the tiling argument in §8.

518 **x_cell** (hexagon) — one mode-0 d_cell joined with one mode-1 d_cell on their matching c2
 519 edge. This is the public cell of the Hex9 grid: 12 root x_cells cover the globe, and each x_cell
 520 has 9 children.

521 **x_dig** — the digit (0–8) naming a child x_cell within its parent. Read it in ternary: the high
 522 trit encodes mode ownership (0 = the child's mode-0 half is interior to the parent context; 1 =
 523 the mode-1 half is interior; 2 = the child is **split**, straddling two parents), and the low trit records
 524 the c2 orientation of the child's long edge. The three split children per parent ($x_dig \in \{6, 7, 8\}$)
 525 are the only cells with two valid parents (§10b).

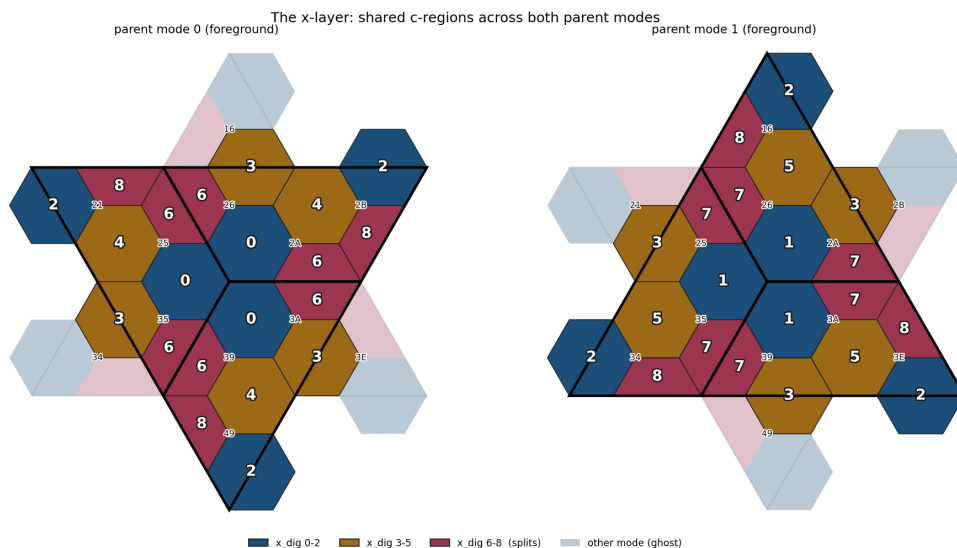


Figure 9: The x-layer: a parent x_cell and its 9 child x_cells labelled by x_dig (0–8), coloured by high-trit class (0–2 / 3–5 / 6–8). The three split children (6,7,8) straddle the parent boundary; there is no central child.¹⁵

526 **Lists and addresses** — c_list, t_list, and d_list are digit sequences over strict single-parent
 527 trees: they compose left-to-right and are prefix-sortable. An **x_list** is the sequence of x_digs;
 528 because of the split cells it is resolved right-to-left. An **x_adr** is an x_list plus a **tail** — a
 529 metadata nibble that resolves the root octant and split-cell parentage (§10b details its fields).
 530 The tail is metadata only: it never participates in geometric computation.

531 10a. Identity as Locator

532 A Hex9 address is a pair — but not the pair a reader might expect. It is not (octant, path); it
 533 is (path, metadata). The path is the x_list: the sequence of x_dig values recording which child
 534 x_cell was entered at each level of the hierarchy. The metadata is the tail. The tail's root-mode
 535 field, together with the level-0 hexagon digit at the head of the path, identifies the octant — the
 536 octant is derived, not stored — and its parent-c2 field resolves the parentage ambiguity that the
 537 split digits 6–8 bake into the path (§10b develops both fields). An x_adr of depth L identifies
 538 exactly one x_cell at level L on the reference ellipsoid.

539 The reversibility of this mapping rests on the construction. At each level, a parent x_cell contains
 540 exactly 9 child x_cells, whose arrangement is determined by the t_cell → d_cell → x_cell

¹³Generated from the `addressing.py` classifier.

¹⁵Generated by `examples/ex0400_anatomy.py`.

sequence established in §§6–8. Each x_dig (0–8) selects one of those 9 children unambiguously. The refinement tree is a strict single-parent structure for t_cells and d_cells ; x_cells inherit this property except at the 3 split x_cells per level, whose parentage is resolved by the tail of the x_adr .

Once the projection from the reference ellipsoid to the octant 2D plane is complete, the encode direction ($point \rightarrow x_adr$) operates via a sequence of linear inequality evaluations. Three families of parallel lines partition the octant plane into the triangular grid:

- horizontal bands: y compared against fixed thresholds;
- positive-slope bands: $y - \sqrt{3}x$ compared against fixed thresholds;
- negative-slope bands: $y + \sqrt{3}x$ compared against fixed thresholds.

A point’s t_cell at a given level is determined by which band it occupies on each family — a small fixed number of comparisons, with no geometric distance computation and no iterative search. At each successive refinement level the thresholds scale by $1/3$, maintaining identical structure. The c_cell (the 96-slot classifier combining horizontal and slope bands, pictured in §10.0) maps directly to a t_cell , and from there to the d_cell and x_cell via the construction of §§6–8.

The mapping operates in both directions. Given an x_adr , the corresponding geographic region is recovered by tracing the digit sequence from the octant root: each x_dig selects a child x_cell whose boundary is determined by the refinement geometry. Given any point on the ellipsoid, its x_adr at level L is recovered by projecting to the octant plane and evaluating the inequality sequence down to depth L .

Both directions are exact at each finite level. Encode produces no approximation: every point belongs to exactly one x_cell at every level. Decode recovers a cell whose geographic extent is precisely determined by the octahedral embedding and the refinement geometry.

This reversibility qualifies the Hex9 address as a geocode, and Hex9 itself as a locating system rather than merely an indexing scheme. Location is not inferred from the address — it is recovered from it by retracing the construction.

10b. Identity as Key

A Hex9 address is also a spatial key — a string over a small digit alphabet that can be stored, compared, and sorted without reference to any geometric structure. This makes Hex9 addresses directly usable as database keys, hash keys, or binning primitives.

The key property is that prefix order corresponds to containment — stated with the care that Hex9’s split cells demand. For the strict single-parent layers the property is unqualified: every c_list , t_list , or d_list beginning with a prefix σ names a cell inside the cell σ names. At the x layer, three facts coexist. First, a canonical body is a unique name even without its tail: no two cells at the same level share a digit body anywhere in the canonical address space — machine-verified exhaustively through level 5 (12, 108, 972, 8,748, 78,732 and 708,588 bodies; all distinct).¹⁶ Second, a body is nevertheless a *lineage* path (§10c), and lineage is not ancestry: cutting a canonical body at length K yields its lineage node at level K , which for a child in the mode-1 half of a split parent is not its canonical ancestor. In the London example below, 43585 cuts to 4358, but its canonical level-3 ancestor is the hexagon canonically named 4348 — 4358 is that same hexagon’s reading under its non-canonical mode-1 parent, while the body 4358 canonically names a different hexagon elsewhere. Prefix comparison therefore groups by lineage; grouping by canonical ancestor derives the ancestor through the tail first (the caution below). Third, geometry: a split child straddles its two parents — only its mode-0 half lies

¹⁶Verified by `experimental/body_census.py`: every cell at every level 0–5 is enumerated and the tail-stripped bodies compared. Non-canonical readings of a body do denote other cells — the tail pins the reading — but the canonical pipeline never emits them.

inside its canonical parent — and exact nesting lives one layer down, at the `d_cell` (§10e). Prefix containment is not the claim that every child hexagon sits wholly inside a parent hexagon; no hexagonal hierarchy can claim that, since hexagons do not tile hexagons.

The operational consequences hold with the lineage/ancestry distinction kept in view. Cells belonging to a given level-K region are found by canonical truncation — a tail-aware cut, implemented as `h9_bin` for full-depth addresses and `h9_cell_ancestor` for cell keys (§13c) — after which containment and join queries reduce to exact key and prefix comparison, with no geometric computation: two observations share a cell at level K if and only if their canonical level-K bins agree. Raw string cutting remains meaningful — it groups by lineage, and away from split lineages the two groupings coincide — but at split lineages they differ, which is the London table’s lesson. Aggregation across levels truncates canonically for the same reason. These operations are efficient and require no coordinate arithmetic.

Because the address space is defined by the refinement structure rather than by a numerical coordinate grid, there are no edge effects and no wrap-around anomalies of an imposed coordinate system. Every cell has exactly one canonical address; every canonical address identifies exactly one cell. The key space is clean.

The tail is what grounds these operations — not for uniqueness of canonical names, which the census above shows the bodies already have, but for *reading*. A digit sequence in isolation does not carry its parent context: 3 of the 9 child `x_cells` at each level ($x_dig \in \{6, 7, 8\}$) straddle the `c2` boundary between two adjacent `d_cells` and have two valid parents, and the same terminal digit can arise from different parent `c2` contexts. The context that makes a body’s reading unique is global — recoverable in principle by re-descending the hierarchy, but not from the digits locally. The tail pins the reading in place, making decode a local, constant-time operation.

The key tail carries two fields. The first is `p_c2` — the parent `c2` of the terminal region. (Within a parent context, a child `t_cell` occupies one of 12 positional classes, its **region** id, 0–11: 6 classes are shared across both parent modes, 3 occur only under a mode-0 parent, and 3 only under a mode-1 parent.) A concrete instance: terminal digit 6 arises both from region 9 (mode 1, parent `c2` = 0) and from region 6 (mode 0, parent `c2` = 2). The two readings denote different cells; only one is the canonical reading of any given body, but without `p_c2` the decoder cannot tell which without global re-descent. The second field is `r_mo` — the root octant’s `net_mode`. The root digit names one of the 12 root hexagons, each of which spans two octants; `r_mo` selects the octant frame in which the descent is read, without which the path cannot be oriented geometrically.

The third field is `p_mo`, the mode of the terminal region’s parent. Its role is resolved by canonicalisation: encoding folds the terminal half-hex onto its mode-0 parent *before* the tail is written, so every canonical address carries `p_mo` = 0. The bit exists because a mode-1-parent reading arises naturally during construction — split cells have two valid parents (§10.0) — and must remain expressible, and the decoder honours it; but the canonical pipeline never emits one. With `p_mo` pinned to zero, the reversible tail and the key tail coincide: a single nibble, identical in both styles.

The tail fields in summary:

Field	Bits	Carries	Without it
<code>p_c2</code>	2 (bits 2–1)	parent <code>c2</code> of the terminal region	decode ceases to be local: the canonical reading of the terminal digit is recoverable only by global re-descent

Field	Bits	Carries	Without it
r_mo	1 (bit 0)	root octant net mode	octant frame ambiguous — each root hexagon spans two octants; the path cannot be oriented geometrically
p_mo	1 (bit 3)	parent mode of the terminal region; 0 in every canonical address	non-canonical (mode-1-parent) readings inexpressible during construction

The tail is invertible in the sense that matters: digit body plus tail round-trips to the canonical cell — its region and its representative point — never to the original source point, which quantisation necessarily discards. Reversibility is thus bounded by resolution, not by the encoding: at full UUID depth the terminal cell is roughly 40 nm across (§13b), so the round-trip is exact for any geodetic purpose, while at coarser bins the recovered representative and the source point can differ by up to a cell radius — by design. §13b details the packed form and the marker that distinguishes a bin from a full-depth address.

One caution follows directly. Truncating an address to length K always identifies a valid ancestor at the corresponding level, but not necessarily the *canonical* one: a split cell encoded under its mode-1 parent truncates into the mode-1 lineage. Binning by naive prefix-cutting therefore silently produces two bins for the same cell whenever non-canonical addresses are present. The correct operation derives the canonical ancestor via the tail before truncating (`h9_cell_parent` and its composition `h9_cell_ancestor`; §13c): prefix-cutting is exact for resolution identification; canonical ancestry requires the tail.

A worked example: central London. The Prime Meridian is a c2 boundary in this region. At level 4, the cells around Greenwich sit in three adjacent hexagons with non-adjacent prefixes:

L4 digit body (lineage)	Location
43483	corner of north London
43486	east of Greenwich
43527	west of Greenwich

43527 appears to have jumped lineage: it belongs to 4352 (south-west England) despite being geographically adjacent to 43486 in 4348 (east England). The jump is not an anomaly — it is the visible signature of the split digits. Digits 6–8 carry high ternary trit 2: these cells straddle the d_cell boundary, and the canonical mode-0 parent convention places geographically adjacent cells on opposite sides of that boundary into different canonical lineages, exactly as the construction requires.

The deeper form of the jump is 43585 (see the figure). Its terminal digit is interior — nothing about the cell itself straddles — yet it sits wholly inside the heavy 4348 outline while carrying the 4358 lineage. The reason is that 4348 is itself a split cell: it straddles the [434|435] level-2 boundary that the Prime Meridian follows here, and its canonical name descends from its mode-0 parent, 434. The children of its mode-0 half extend 4348; the children of its mode-1 half descend through 435 and extend 4358 — one hexagon, two lineage prefixes — while canonically the body 4358 names a different split hexagon elsewhere, the one whose mode-0 parent is 435. This is also the mechanism behind the census result above: canonical bodies never collide because each cell’s canonical name claims exactly one (parent, digit) slot, but the lineage prefixes visible on a map are not canonical ancestors, and aggregation must derive ancestry through the tail.

¹⁸Rendered in QGIS over an OpenStreetMap backdrop; hex boundaries and addresses from `hhg9`.

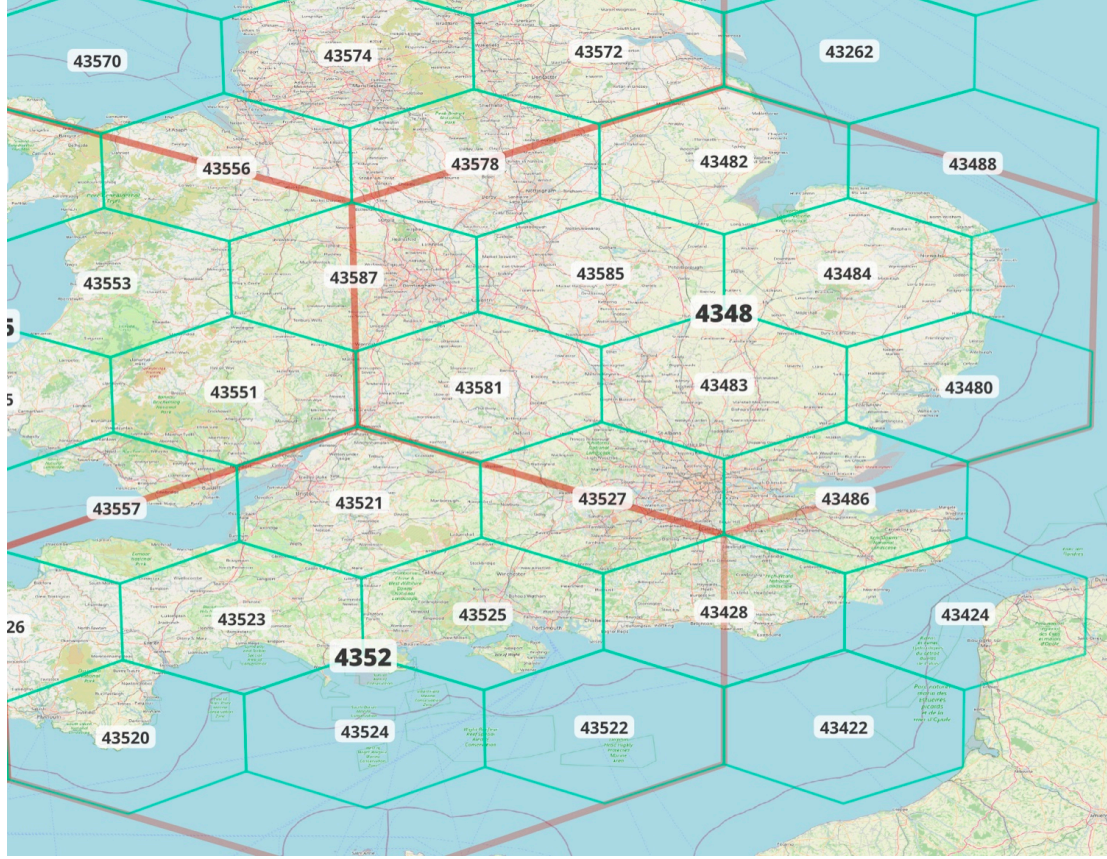


Figure 10: Split-cell lineage structure over England (Equal Earth, EPSG:8857). Hexagons are annotated by digit body — the lineage path, §10c; the bold **4348** and **4352** mark two level-3 cells and the heavy lines their boundaries. Two forms of lineage jump are visible. Cell **43527** lies west of Greenwich yet belongs to the **4352** lineage while its geographic neighbours **43486** and **43483** belong to **4348**: a split child, straddling the boundary, canonically named from across it. Deeper: 4348 is itself a split cell on the [434|435] level-2 boundary that the Prime Meridian follows here. Its canonical name descends from 434, but the children of its mode-1 half — **43581** and **43585**, wholly interior to the heavy 4348 outline, plus the split child **43587** on its rim — descend through 435 and carry the 4358 lineage: one hexagon, two lineage prefixes. Canonically, the body 4358 names a different hexagon elsewhere; canonical bodies themselves never collide (machine-verified, §10b).¹⁸

10c. Identity as Label

A label is the human-readable serialisation of the key: the digit body followed by the key-tail nibble, written `<body>.<tail>` — 32343.2 names the level-4 cell whose canonical digit path is 32343 under key tail 2. The form is implemented by `h9_label` and its inverse `h9_from_label`; labels and keys are in bijection, so a label carries exactly the identity of §10b, in a form that can be printed, sorted, and compared by eye.

The digit body alone — the part before the dot — is a *lineage path*, not a decodable name. It is readable and prefix-meaningful: truncating it walks up a lineage, and two bodies sharing a prefix share that lineage context. The census of §10b even shows canonical bodies never collide — yet the body still falls short of a name in two ways. First, lineage is not ancestry: cutting a canonical body yields lineage spellings, and at a split parent the spelling differs from the canonical name of the cell it reaches (43585 cuts to 4358; that hexagon’s canonical name is 4348). Second, a body cannot be read locally: the same digit string under different tail readings denotes different cells, so recovering a bare body’s cell requires global re-descent (§10b). The tail nibble closes both gaps. A string without it should be read as a path through the hierarchy, not inverted as a cell.

What makes labels uniform is the canonical fold of §10b, not a suppression of structure: after canonicalisation every cell is presented through a mode-0 terminal parent, so a label needs no mode flag, lookup table, or supplementary field to interpret. The bipartite mode structure remains present in the refinement geometry and is recoverable from the key. A label therefore identifies a specific geographic region at a specific resolution, in one canonical spelling — nothing more is needed, and nothing is omitted.

10d. Adjacency from Refinement Paths

Two `x_cell`s are adjacent if they share a boundary edge on the reference ellipsoid. In Hex9, adjacency is recoverable from the refinement structure because the `x_cell` geometry is fully determined by the `t_cell` → `d_cell` → `x_cell` construction.

Within a parent `x_cell`, the tiling one layer down is exact — but it is a tiling by `d_cells`, not by nine whole hexagons. The parent region is covered by its 6 interior child `x_cells` (digits 0–5, twelve connected `d_cells`) together with 6 rim `d_cells`, each the inward-facing half of a split hexagon. Three of those rim `d_cells` are the mode-0 halves of the parent’s own canonical split children (digits 6–8); the other three are mode-1 halves of split cells canonically owned by the parent’s neighbours (§10b). Shared edges throughout are the `c2` edges of the underlying `d_cells` — the same long-edge alignment that the orientation selection of §8 fixed globally.

Adjacency is computed at the `t_cell` layer, where it is a fixed, finite lookup: every `t_cell` has exactly three neighbours, one per `c2` edge, given by a constant table keyed by (region id, parent mode, `c2`). The digit structure mirrors the assembly: each split digit `k+6` ($k \in \{0, 1, 2\}$) names the pair of `d_cells` flanking interior child `k`, and the low ternary trit of every `x_dig` records the `c2` orientation of the cell’s long edge. `d_cell` and `x_cell` adjacency compose from this `t_cell` table through the `t` → `d` → `x` assembly: the six neighbours of an `x_cell` are reached through the `c2` edges of its two halves.

The same `t_cell` table classifies every child’s relationship to its parent `t_cell`’s boundary — the figure above shows the classes. Within each parent `t_cell`, child `t_cells` fall into three classes by their count of parent-crossing edges: interior children, whose three neighbours all share the parent; mid-edge children, with exactly one neighbour in an adjacent parent; and vertex children, with two. For the latter classes the lookup flags the crossing and identifies the corresponding child in the neighbouring parent — a computable operation on the address structure, stepping up one level in the refinement tree and applying the same `c2` edge relationships that govern interior

²⁰Generated by `examples/ex0400_anatomy.py`; classes verified against `region.py`’s neighbour builder.

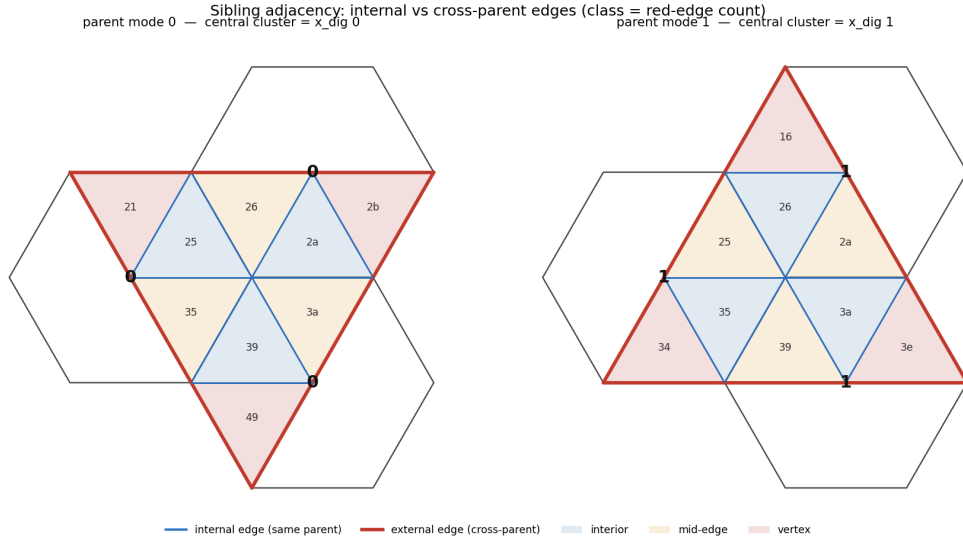


Figure 11: Sibling adjacency within a parent: each child t_cell edge is internal (blue, shared with a sibling) or external (red, cross-parent); a child’s adjacency class is its red-edge count (interior 0, mid-edge 1, vertex 2).²⁰

adjacency. Where the neighbouring parent lies across an octant seam, the octant congruence of §5 reduces the hop to an octant-index lookup composed with the y-reflection.

Adjacency across octant seams follows from the d_cell $c2$ alignment at octant boundaries. The orientation selection of §8 ensures that d_cells at an octant edge meet long-edge to long-edge with the d_cells of the neighbouring octant. The x_cells that form across this boundary are assembled by the same d_cell join rule as everywhere else. Seam-crossing adjacency is therefore not a special case — it is the same $c2$ -edge query applied at the octant boundary.

In each case adjacency is a finite computation on the combinatorial structure of the refinement tree and the $c2$ edge table. No geometric distance query is needed, and no location in the grid requires a different procedure from any other.

10e. Continuity

At any finite level L , a Hex9 address identifies a cell of finite geographic extent. As L increases, cell diameter decreases by a factor of 3 at each level, converging toward zero. An infinite address sequence — a path carried to all levels of the hierarchy — therefore defines a nested sequence of cells whose diameters tend to zero.

The reference ellipsoid is a compact metric space. By Cantor’s intersection theorem, a nested sequence of closed regions with diameters converging to zero has exactly one point in its intersection. An infinite Hex9 address sequence identifies exactly that point: not a region, but a location on the ellipsoid.

This convergence lets a function recover position from an address to arbitrary precision: decoding an address of growing length yields a sequence of points converging to a unique location on the ellipsoid (Appendix A). A finite address identifies a region; a sufficiently long address identifies a location to any required tolerance.

This is not the same as being a coordinate reference system in the strict sense of ISO 19111, and we do not claim it is. Two limitations are intrinsic. First, the address alphabet is discrete: addresses form a totally disconnected sequence space, not a continuum, so they are not coordinates in the real-valued sense. Second, the point $\rightarrow$ address map is discontinuous on the measure-zero set

of `d_cell` seams (the split-cell boundaries of §10b): arbitrarily close points on opposite sides of a seam receive addresses that differ in their leading digits. ISO 19111 presupposes continuous coordinates, and Hex9 does not meet that requirement.

What Hex9 offers is better described, by analogy with its quasi-authalic geometry, as **quasi-continuous**: position is recoverable from the address by a function, to arbitrary precision, everywhere except on a measure-zero seam set, and the cell hierarchy converges to points rather than terminating at a finite floor. We are not aware of another DGGS whose zonal identifier doubles as a position-recovery coordinate in this way, though we do not claim the property is unique. At any fixed finite resolution Hex9 remains a discrete system — cells are regions, the bijection is between addresses and regions — and the quasi-continuous behaviour emerges only in the limit. This is consistent with §9: Hex9 is a discrete spatial reference system that approaches, but does not attain, a continuous one.

10f. Seams and Valence Defects

As stated in §4, the Euler characteristic of S^2 requires that any triangulation of the globe carry topological defects. In Hex9 these are absorbed at the six octahedral vertices — each a 4-valent point of the triangulation, where the hexagonal grid carries its defect as the five-neighbour cells of §12. Understanding why neither these defects nor octant seams require special handling requires tracing the constructive sequence: $t_cells \rightarrow d_cells \rightarrow x_cells$.

At each refinement level, each triangular face (`t_cell`) is subdivided into 9 child `t_cells`. Three adjacent `t_cells` — grouped by their shared long edge (`c2` edge) — form a half-hexagon (`d_cell`). A `d_cell` carries an intrinsic mode (0 or 1). One mode-0 `d_cell` and one mode-1 `d_cell`, joined on their matching `c2` edge, form a hexagonal cell (`x_cell`). The hexagonal grid emerges entirely from this sequence; no independent hex tiling is assumed.

(The $t_cell \rightarrow d_cell \rightarrow x_cell$ constructive sequence is the anatomy figure of §10.0.)

At an octant seam, the `d_cell` `c2` alignment ensures that `d_cells` on either side of the shared octant edge meet long-edge to long-edge. This alignment is not enforced separately — it is the consequence of the orientation selection of §8, which chose precisely the arrangement in which `c2` edges align at every boundary. The `x_cells` that straddle the seam are formed by the same `d_cell` joining rule that applies everywhere else. The seam is a boundary in the refinement tree, not a discontinuity in the construction.

At an octahedral vertex, four octant faces meet — the six seed vertices are 4-valent, where every interior vertex of the refined triangulation is 6-valent. The surrounding `d_cells` at this vertex have `c2=0` — the flat (horizontal) edge — as their shared long edge. This is the `c2=0` convention: not a patch, but the consequence of the same vertex-loop transport closure established in §4. Even valence at the octahedral vertices (valence 4) satisfies the mode transport condition; the surrounding `x_cells` are formed by the same `d_cell` join rule. The cells meeting at the vertex are the five-neighbour hexagons of §12, structurally identical in construction to every other `x_cell` — each a hexagon that, projected onto the ellipsoid, presents one collinear corner and so appears pentagonal.

In both cases the construction proceeds without branching. The $t_cell \rightarrow d_cell \rightarrow x_cell$ sequence applies uniformly across the entire globe — at seams, at defect vertices, and in the interior. The absence of special cases follows from the coherence of the construction, not from exception handling added afterward.

11. Geometric Realisation (AK + Warp)

Sections 1–10 establish the Hex9 grid as a combinatorial object: mode transport, `t_cells`, `d_cells`, `x_cells`, the refinement hierarchy, and the address structure are all defined without reference to any specific map projection or reference body. The geometric realisation is a separable step that places this abstract structure onto the WGS84 reference ellipsoid.

Three concerns are independent: (1) the combinatorial structure and hierarchy (§§1–10); (2) the base projection from octahedron to ellipsoid; (3) the area correction. Each can be understood, improved, or substituted without disturbing the others. The warp is ellipsoid-specific; the grid is not.

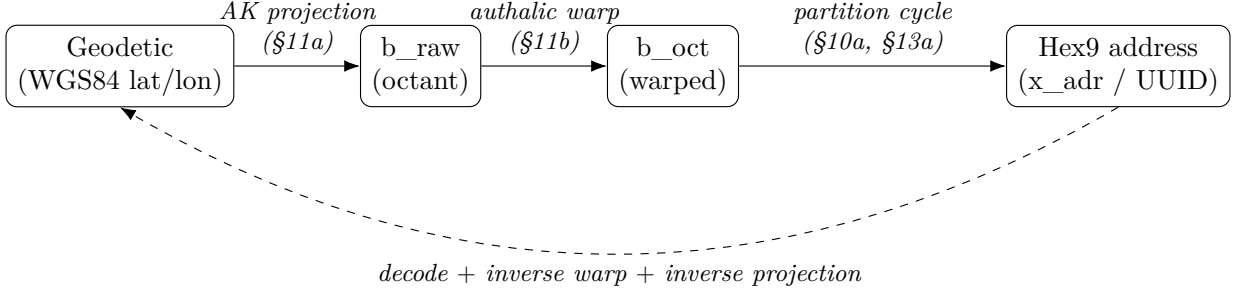


Figure 12: The geometric-realisation pipeline (§11). A WGS84 geodetic coordinate reaches a combinatorial Hex9 address through the AK base projection, the authalic warp, and the partition cycle; the inverse (dashed) retraces the chain: decode, a Newton–Raphson inversion of the warp (§11b), and a guarded Gauss–Newton inversion of the base projection (§13f). The three concerns — combinatorial hierarchy, base projection, area correction — are independent and separately substitutable.

11a. The Base Projection

The AK octahedral projection maps each of the 8 octant faces to the corresponding region of the reference ellipsoid. Its forward formula, designed analytically by Anders Kaseorg (Kaseorg 2025) from a force-directed dataset, applies a tangent substitution to each octant coordinate and couples the three axes with a fourth-root term modulated by a parameter $\alpha \approx 3.2278$. This coupling partially compensates for the area asymmetry between mode-0 and mode-1 triangular cells that arises from the non-uniform Jacobian of the octant face.

The projection is smooth and has an analytical Jacobian — properties that make it suitable as the inner layer beneath the Sinkhorn warp. The 8 octahedral vertices coincide exactly with ellipsoidal surface points; no projection computation is needed at those locations. The 8 octant faces are mutually equivalent under a y-coordinate reflection, so one projection function serves all octants.

No closed-form inverse exists. The backward pass — ellipsoid to octant — is realised by numerical root-finding; the fast path is a guarded Gauss–Newton iteration on the analytic Jacobian (§13f),²¹ with an exact beam search as the reference and as the fallback near seams and vertices. The forward map is smooth and injective on each octant, so an inverse is guaranteed to exist; the numerical method is an implementation choice, not a structural requirement.

²¹Two Newton-family solvers appear in the realisation, and they are not interchangeable: this projection inverse is a Gauss–Newton least-squares iteration — three ECEF residual components over two face coordinates — while the warp inverse of §11b is a plain Newton–Raphson step on a square 2×2 system. The two methods coincide exactly when the residual system is square with an invertible Jacobian; each use in this paper names its own component.

Without further correction, the AK projection introduces roughly $\pm 20\%$ area deviation across the octant face — larger apparent cells near octant corners, smaller near the centre.

A second, subtler artefact is an inter-mode area bias. The octant face is a right isosceles triangle, so its three corners are not geometrically equivalent and the AK Jacobian is not constant over the face; mode-0 and mode-1 triangles sample that non-uniform field at systematically different centroid positions. Measured by geodesic area on WGS84, the mode means differ by 0.394% at L4 and 0.131% at L5, shrinking roughly threefold per refinement level (projected $\approx 0.04\%$ at L6). The coupling parameter α is not the cause — removing it ($\alpha = 0$) worsens both the global deviation ($\sigma \approx 7.6\% \rightarrow \approx 17\%$) and the asymmetry — and the Sinkhorn warp, operating at the hexagon level, cannot rebalance areas between the two halves of a hexagon, so the residual carries through to the warped result at slightly reduced magnitude. At L5 and finer it is negligible for practical use.

11b. The Authalic Warp

The warp corrects the area deviation left by the base projection. (“Authalic” means equal-area: the warp is not itself a projection but the corrective deformation that renders the composite octahedral map area-true.) It is derived by Sinkhorn optimal transport (Cuturi 2013): treating the L4 (or L5) triangle vertices as a discrete mass distribution on the octant, the Sinkhorn iteration finds the minimal-displacement redistribution that equalises projected cell areas against the geodesic areas they subtend on WGS84. The result is a displacement field — corrections in `b_oct` coordinates — rather than absolute positions, which improves numerical conditioning.

The displacement field is precomputed once per ellipsoid and stored. At runtime, it is applied via a Clough-Tocher interpolant (Clough and Tocher 1965) (C1 continuous). The inverse warp uses the forward interpolant to obtain an initial estimate, then refines by Newton–Raphson to a tolerance of 10^{-14} in `b_oct` (barycentric) units. The geodetic round-trip $g \rightarrow b_oct \rightarrow g$, measured on WGS84 at validation points that include a near-pole location (89.99°N) and the Greenwich seam, returns to within 1.8 nm of the original position — many orders of magnitude below any geodetic relevance, and comfortably under the 7 nm design threshold.

The achieved area uniformity (L5, all 708,588 hexagons, WGS84, production warp file, geodesic areas): mean deviation exactly 0.000% (confirming closure: the cell areas sum to the ellipsoid surface area); area deviation min -3.57% , max $+4.80\%$; mean absolute deviation 0.001%; log-ratio standard deviation 1.99×10^{-4} . Half of all cells are within 0.0002% of ideal area; 99% within 0.0044%; 99.99% within 0.43%. The extreme values are a balanced $\pm 4\%$ pair affecting a very small number of cells immediately adjacent to the six octahedral vertices; the bulk distribution is highly uniform. The warp is quasi-authalic rather than strictly authalic. A strictly authalic projection constrains only the Jacobian determinant, permitting severe shear and cell elongation. The optimal-transport derivation implicitly regularises against shear by minimising displacement: the result trades a small area residual for a smooth, low-distortion displacement field.

Six irreducible vertex singularities remain at the octahedral poles — the geographic N and S poles and the four equatorial points at 0° , 90° , 180° , 270° E. This residual is topological in origin: the Gauss-Bonnet theorem requires concentrated curvature at the 4-valent octahedral vertices, and no smooth warp can remove it entirely. The singularities are fixed and geometrically determined. On WGS84 they are not symmetric: the geographic poles produce more pronounced residuals than the equatorial vertices, a consequence of ellipsoidal oblateness concentrating curvature at the minor-axis tips. The census is sharper still: the six vertices fall into *three* pairwise-exact

²³Generated by `examples/ex0081w_warped_authalics.py` (`snow_globe`, `lim_pct=0.25`).

²⁵Generated by `examples/ex0118_mollweide.py` (L5).

²⁷Generated by `examples/ex0124_vertex_census.py` (census table and figure; shares the cached geometry of `ex0081w_warped_authalics.py`).

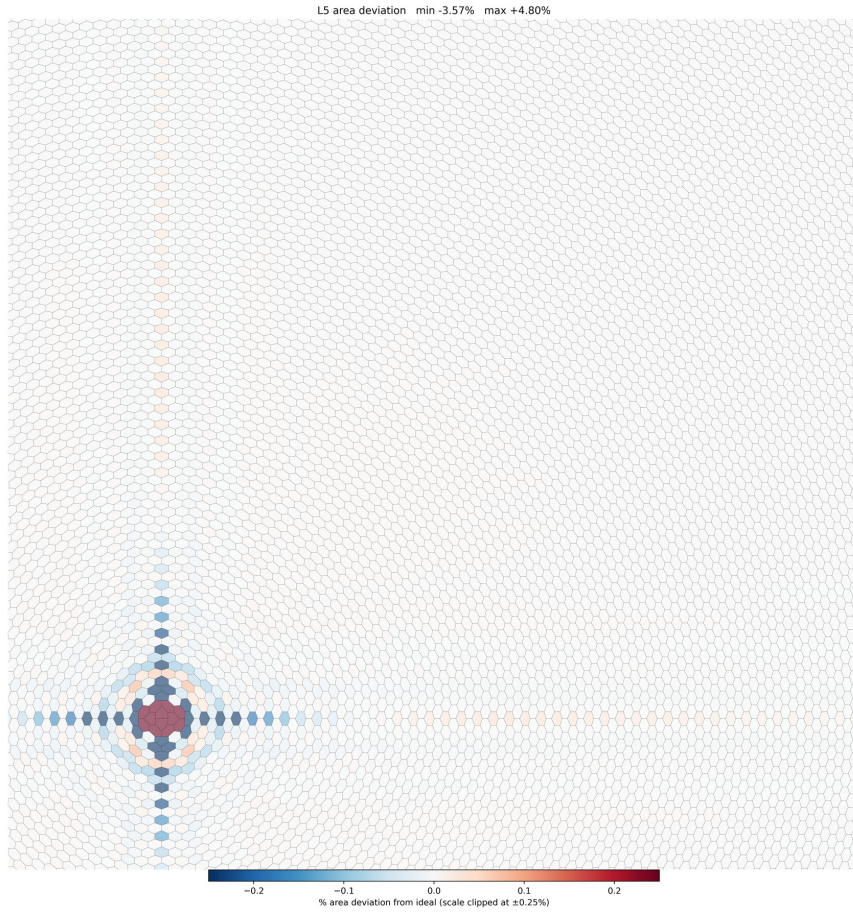


Figure 13: Per-hex area deviation around the north-pole vertex, colour scale capped at $\pm 0.25\%$. The tight cap reveals the residual structure: a quiet near-white interior (the equal-area result), the vertex bloom whose innermost cells reach the $\pm 4\text{--}5\%$ extremes and saturate the scale, and the gentle grading that runs out along the two visible octant seams. Every cell is one L5 hexagon.²³

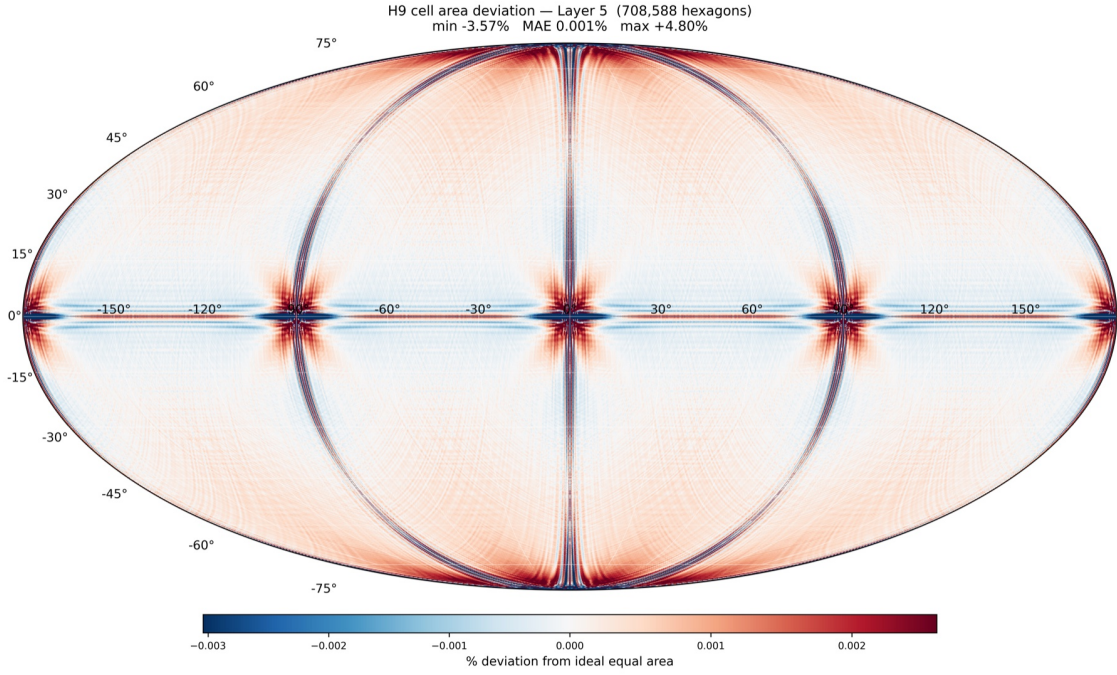


Figure 14: Per-hex area deviation, Mollweide, on a scale clipped at $-0.31\%/+0.26\%$ — the p1–p99 band (pattern view). The clip makes the spatial structure legible: a quiet near-white interior with the octant seam skeleton and six vertex blooms picked out. The colourbar is clipped precisely so that this structure shows: unclipped, the vertex blooms would own the whole scale and the field would wash out to near-uniform white everywhere else. It therefore does **not** reach the true extremes (min -3.57% , max $+4.80\%$).²⁵

classes — the poles, the $0^\circ/180^\circ$ pair, and the $90^\circ\text{E}/\text{W}$ pair — distinct in count, extreme, sign, and morphology (figure). The polar/equatorial split follows from oblateness; the split between the two equatorial pairs cannot (the ellipsoid is rotationally symmetric) and is structural in origin — the pairwise-identical spectra rule out numerical noise. The structure is visible at L0: exactly two root cells meet at each octahedral vertex (the seed solid, §9), and the axis of their join takes three orientations against the graticule — diagonal at the poles, meridional at $0^\circ/180^\circ$, equatorial at $90^\circ\text{E}/\text{W}$. A control run with the warp disabled confirms the seed: the raw AK realisation separates the same three classes with the same pairwise exactness, though minutely (class differences of order 0.005% inside its $\pm 20\%$ field); the warp, in removing the shared bulk deviation, leaves a class-dependent residual that magnifies the equatorial-pair separation roughly a hundredfold, to the $\approx 0.5\%$ shown.²⁸

11c. The Native Space

After the warp, each x_{cell} occupies a well-defined region in b_{oct} — the octant barycentric space.²⁹ In b_{oct} , the hexagonal cells are congruent and regular: no cell is larger, smaller, or differently shaped than any other (modulo the irreducible vertex residuals). b_{oct} is the natural coordinate space for Hex9 computation; it is where the inequality evaluations of §10a operate and where address digits correspond to regular geometric subdivisions.

Much of the shape variation that appears when Hex9 cells are rendered in a conventional map

²⁸Control run: `examples/ex0124_vertex_census.py --raw` (warp disabled via `b_oct.no_warp()`); raw AK leaves only 4,440 of the 708,588 cells within 0.1% of ideal area — the warp brings that to all but 244.

²⁹The name is historical: b_{oct} coordinates are planar Cartesian coordinates on the unfolded octant net, measured in octant-normalised units — not simplex (barycentric-weight) coordinates. No convex-combination reading is intended.

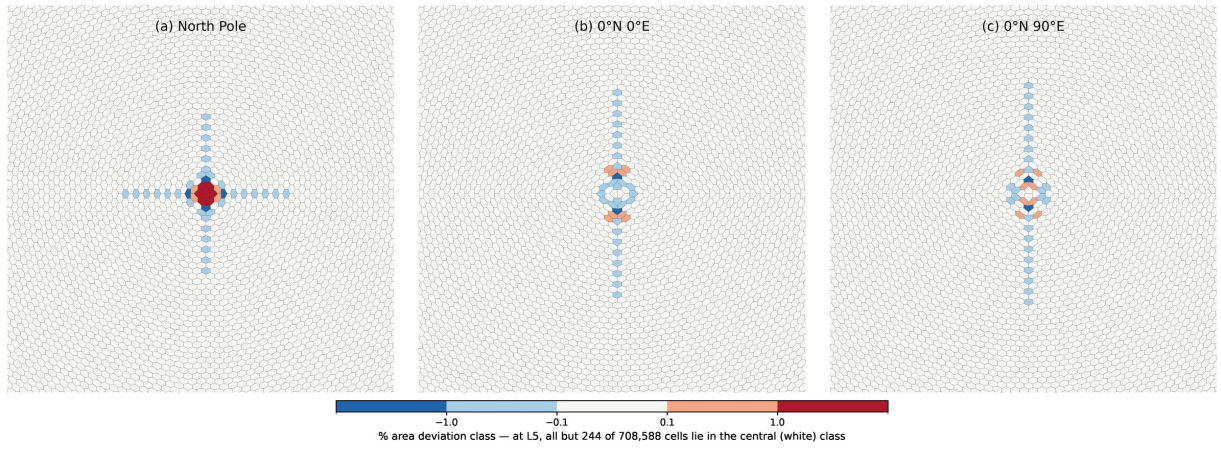


Figure 15: Where the “quasi” in quasi-authalic lives: every L5 cell classified by signed area deviation, showing one vertex from each of the three vertex classes at a common scale — (a) North Pole, (b) 0°N 0°E, (c) 0°N 90°E; every hexagon is one L5 cell ($\approx 720 \text{ km}^2$, $\approx 33 \text{ km}$ across). All but 244 of the 708,588 cells lie in the white class (within 0.1% of ideal area). The exceptions cluster at the six octahedral vertices in three pairwise-exact classes: the poles (48 cells beyond 0.1% each, 12 beyond 1%, extremes $-3.57\%/+4.80\%$), the 0°/180° pair (38 and 2, extreme -2.10%), and the 90°E/W pair (36 and 2, extreme -2.60%). Within each pair the sorted deviation spectra agree beyond three decimals, so the three classes are structural, not numerical. Morphology follows class: the polar bloom carries an excess (red) core with rays on all four meeting seams; the equatorial blooms have near-tolerance cores with their extremes wholly in deficit (blue), and their rays are meridional only — the equator seams stay within tolerance throughout.²⁷

projection — elongated cells near the poles in Mercator, apparent distortion in geographic lat/lon — is a consequence of that reprojection, not a property of the cells: in their native `b_oct` space the cells are regular by construction, and that added distortion belongs to the map. This is not to claim the cells are distortion-free on the ellipsoid itself — laying an *equal-area* hexagonal grid on a curved surface costs some bounded shape distortion (the octahedral and warp distortion of §12, aspect ≈ 1.37), and that part is intrinsic. The narrower point holds: a conventional projection *adds* its own distortion on top, and that addition — not the grid — is what a Mercator or lat/lon plot mostly shows.

This is the practical meaning of the claim in §9 that Hex9 is a spatial reference system rather than merely an indexing scheme: `b_oct` is a legitimate projected coordinate space in its own right, realised by the AK+Warp projection (§13e). Conventional geographic renderings are derived from it, not the other way around.

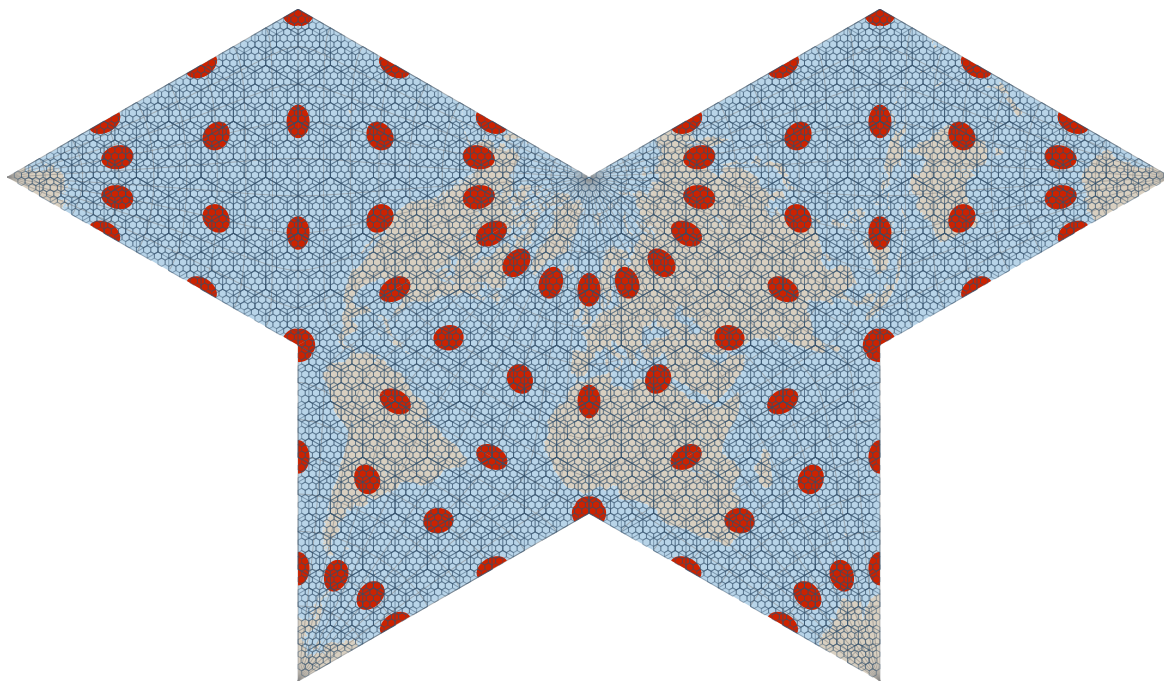


Figure 16: Tissot indicatrices on the AK+Warp `b_oct` butterfly net: near-constant-radius circles across the whole net demonstrate equal area, and their near-circularity the low shear the optimal-transport derivation buys (§11b). For the same grid seen through conventional projections — where the added distortion belongs to the target projection, not the grid — see the §14c comparison panels.³¹

11d. Projection Independence

The AK+Warp combination is the default realisation of the Hex9 grid on WGS84, chosen because quasi-equal-area cells make point counts and density estimates geographically comparable. It is not the only valid realisation.

The combinatorial Hex9 structure is projection-independent. Any projection from the octahedron to the ellipsoid can be composed with the addressing scheme to produce a valid Hex9 coordinate system. The Lee conformal projection (Lee 1965) produces angle-preserving cells at the cost of non-uniform area; the Snyder octahedral equal-area projection (Snyder 1992) achieves strict $\det(J) = 1$ analytically and offers a closed-form inverse — a statement made here on its published properties: we have not exercised the composition, no working implementation of the octahedral

³¹Generated by `examples/ex0121_tissot_svg.py`.

Snyder having been found to test against. The compositional claim is structural rather than empirical in any case: cell identity, digit assignment, and adjacency structure are unchanged across projection choices.

The warp is also ellipsoid-specific but not ellipsoid-exclusive. Recomputing the Sinkhorn displacement field against a different reference body (GRS80, Bessel, a planetary ellipsoid) yields a Hex9 realisation for that body. The addressing hierarchy and all combinatorial structure carry over unchanged; only the area-correction data changes. WGS84 and GRS80 are sufficiently close that a single warp file serves both in practice — the ellipsoidal difference is negligible relative to the warp residual.

Projection interchangeability is a consequence of the design philosophy: the warp corrects for a specific ellipsoid, but the grid it corrects is defined independently of any ellipsoid. This separation is not an accident of implementation — it is the consequence of insisting that topology, hierarchy, and area correction be treated as distinct concerns from the outset.

12. Comparison with Prior Art

The differences between the established systems — H3, S2, HEALPix, A5 — are well documented in the literature; this section makes only the comparisons that bear on Hex9’s claims.

Property	H3	S2	HEALPix	A5	Hex9
Base polyhedron	Icosahedron	Cube	(sphere-native)	Dodecahedron	Octahedron
Cell shape	Hex (+ 12 pentagons)	Quadrilateral	Mixed quad	Pentagon	Hexagon (no exception cells)
Aperture	7	4	4	5 then 4	9 (shifted)
Equal area	No	No	Yes (strict)	Yes	Quasi (p99 < 0.005%)
Strict ancestry at all levels	No	Yes	Yes	—	Half-hex: yes; hex: by convention
Adjacency isotropy (combinatorial)	Yes	No ($\sqrt{2}$)	Yes	No (elongated)	Yes
Dual DGGS/CRS	No	No	No	No	Yes (quasi-CRS)
Reference body	Sphere	Sphere	Sphere	Ellipsoid	Any ellipsoid (per-body warp; WGS84 + sphere trained)

Exception cells. Euler’s theorem requires exactly 12 units of topological defect — twelve cells short of a full six neighbours — in any hexagonal tiling of the sphere, at every refinement level, independent of resolution. No hexagonal DGGS escapes this; the systems differ in where the obligation lands. H3 pays it as genuine pentagon cells: 12 of them at the icosahedral vertices, first-class exceptions with five children instead of seven, a distinct geometry, and an `is_pentagon()` guard that every correct H3 implementation must carry — the price of odd vertex valence, where a 5-valent vertex cannot sustain the face two-colouring a consistent hexagonal subdivision depends on (§4). Hex9 introduces no exceptional cell type at all: every cell is a hexagon. The same obligation appears instead as twelve five-neighbour hexagons, two at each of the six octahedral vertices, where two of a hexagon’s six sides fold together across the seam to meet the same opposing hexagon. They carry ordinary addresses, are constructed by the same `d_cell` join rule as every other cell (§10f), and require no API guard. On the flat net they are plain hexagons;

projected onto a reference ellipsoid one of their six corners falls collinear, so they render as pentagons while remaining hexagons combinatorially. In Hex9’s native planar net the six vertices lie on the boundary of the coordinate space, so these singular cells straddle the edge rather than appearing as interior anomalies.

Ancestry. The strictly nested unit in Hex9 is the half-hexagon (`d_cell`), not the hexagon. The `d_cell` hierarchy is a single-parent tree: a `d_cell` address composes left-to-right, prefix-truncation always yields its unique ancestor, and no edge of the finest-level half-hexagon tiling crosses a coarser half-hexagon boundary. Hexagons are assembled from two half-hexagons, and 3 of the 9 child hexagons per level (digits 6–8) straddle a `d_cell` boundary and have two valid parents; the canonical mode-0 convention selects one, and the exact ancestor is recovered from the address tail (§10b). Under that convention the canonical parent function is well-defined and deterministic at every level, and roll-up at the `d_cell` level is geometrically exact — but, unlike the unconditional quad hierarchies of S2 and HEALPix, hexagon roll-up requires deriving the canonical ancestor (via the tail) before truncating, and the split-cell ambiguity can nest: a run of split digits stays ambiguous until the tail resolves it. H3’s aperture-7 subdivision is exact in the index — every cell has one deterministic parent, and parents compose transitively — but it bears no consistent relation to geometric containment, and the divergence compounds with depth. Of a parent’s seven children only the central one lies inside it; a generation further down, boundary grandchildren straddle the parent edge roughly half-and-half; another generation down, index-descendants appear that lie entirely outside the ancestor whose address they extend.³² The systems trade off differently rather than one being weaker: Hex9’s half-hexagon hierarchy is exactly nested in area, while its hexagon hierarchy carries a genuine split-cell ambiguity resolved by the mode-0 convention through the tail; H3’s hierarchy is an exact index laid over grids that never geometrically refine one another.

The comparison can be made quantitative without privileging any system’s vocabulary. Every hierarchical DGGS offers two routes from a point p to a coarse cell: bin directly, $\text{cell}(p, L)$, or bin fine and truncate, $\text{ancestor}_L(\text{cell}(p, L+k))$. In fairness, for single-layer binning the direct route is available and correct in every system, H3 and A5 included; the two routes are distinguished only when layers must be *related* — roll-up, drill-down, joining datasets binned at different resolutions — which is the work a hierarchy exists to do. Measured over 10,000 uniformly distributed points, each system through its own native operations and with no polygon geometry involved,³³ the two routes disagree in H3 for a near-constant $\approx 6.5\%$ of points at every depth $k = 1\dots 6$ — the fixed mismatch between the parent hexagon and the Gosper-island-like limit shape of its descendant set — and in A5 for a constant $\approx 35\%$ (the rhombus limit of its pentagon refinement). S2’s routes agree everywhere. Hex9’s hexagon routes disagree exactly on the split-cell band, and the rate decays threefold per level — 16.8% at $k = 1$, 0.07% at $k = 6$, following the predicted $(1/6) \cdot 3^{-(1-k)}$ — because the limit shape of a Hex9 cell’s descendant set is the ancestor itself: the disagreement is a boundary band that thins with depth, not a shape mismatch that persists. Every disagreeing point is assigned to a cell that shares the straddling hexagon with its geometric parent — an adjacent cell, never a distant one — and at the `d_cell` level the rate is zero at every depth. In the workload case — bin once at maximum resolution, roll up to the analysis layer — the measured rates are H3 6.5% ($r_{15} \rightarrow r_5$), A5 35.0% ($r_{30} \rightarrow r_5$), and 0.00% for both S2 ($L_{30} \rightarrow L_8$) and Hex9 ($L_{30} \rightarrow L_5$): a constant rate is structural and depth cannot amortise it, while the Hex9 band decays below any measurable rate ($(1/6) \cdot 3^{-24} \approx 10^{-13}$). For Hex9 that deep case is not a favourable corner but the intended pipeline: a point is encoded once to its full 128-bit address — layer 30, a cell some tens of nanometres across — and every coarser bin is thereafter

³²Demonstrated by `docs/dggs/dggs_nesting.py` (reference `h3`, `s2sphere`, `a5` and `shapely` libraries): the H3 index hierarchy is single-valued and transitive, yet index-descendants become geometrically disjoint from their ancestor by the third generation.

³³`docs/dggs/dggs_commute.py` ($N = 10^4$, fixed seed); binning at each system’s maximum depth (`--deep`) reproduces the constant rates and the Hex9 decay law.

derived from the address alone by truncation and tail canonicalisation, in constant time with no re-encoding from geometry (§13b). In normal use the system therefore operates permanently at the measured-zero end of the table.

This is the signature of any hierarchy built by *iterating* a single-level parent/child relation. H3’s aperture-7 step scales by $\sqrt{7}$ and rotates by $\approx 19.1^\circ$, so each level is turned against the last and the offset accumulates until descendants leave the ancestor outright. A5’s same-prefix refinement behaves the same way — under its own descendant relation its finer cells likewise spill outside the anchor, and a larger fraction of them than H3’s. This is no defect of A5: it is an equal-area index, and its hierarchy preserves area continuity rather than geometric containment — the two cannot be had together on a curved surface, and A5 makes the opposite trade to S2 and Hex9. The systems that nest are the ones whose refinement does not iterate an offset: S2’s quadtree, where squares refine onto squares, and Hex9, whose descendants are not walked down the tree but read directly from a per-layer contract (§7, §10b) — so the split hexes, the one place an iterated step would break, stay bound to their owning ancestor and never decohere. The commutation rates above are the same distinction made quantitative: a constant rate is the signature of a limit-shape mismatch, a geometrically decaying one of an exact limit shape with a thinning straddle band.

Area. HEALPix is strictly equal-area but pays in cell shape: its cells are quadrilaterals of visibly varying geometry. H3 and S2 are not equal-area; cell areas vary by tens of percent across the globe, which biases any analysis that compares counts or densities between regions. Hex9 is quasi-authalic by construction of the warp (§11b): the residual is bounded, characterised, and confined to the neighbourhoods of the six vertex singularities.

Shape and area, independently surveyed. An independent survey of cell geometry — enclosing-cone aspect ratio and WGS84 geodesic cell area over $N = 5000$ uniformly-sampled cells per system (H3, S2, A5, and Hex9; HEALPix is covered qualitatively under *Area*, below) — places the four systems into two area tiers separated by roughly $100\times$.³⁴ A5 and Hex9 form an equal-area tier — A5 exactly equal-area by construction (measured coefficient of variation 0.00%, the residual at the geodesic-area measurement floor), Hex9 equal-area to a coefficient of variation of 0.10% through its corrective warp; H3 and S2 do not (12.5% and 14.5%), and the gap is a step, not a gradient. The survey reproduces Hex9’s warp characterisation independently — worst cell $\approx +4.8\%$, mean absolute deviation 0.001%, all residual pooled at the twelve octahedral-vertex defects — on a third-party tool and against the WGS84 datum, corroborating §11b. On cell shape the order is H3 (aspect ratio 1.06, roundest) $<$ S2 (1.24) $<$ Hex9 (1.37) $<$ A5 (2.14): Hex9 reaches the equal-area tier while staying markedly rounder than the only other equal-area system in the comparison. Both metrics are resolution-invariant (reproduced to four decimals five levels coarser), and the dispersion ranking $A5 < Hex9 < H3 < S2$ is the same under every statistic tested (CV, p90, p95, max/min). Hex9 is therefore the better combined shape-and-area cell among the equal-area systems: A5 attains exact equal area but with markedly elongated cells (aspect 2.14), whereas Hex9 holds equal area to a mean absolute deviation of 0.001% — its residual pooled at the twelve octahedral-vertex defects (max/min 1.059) — while keeping cells far rounder (aspect 1.37).

Aperture. Hex9’s aperture is 9, but the subdivision is not the centred aperture-9 scheme of the DGGS literature. In a centred scheme one child sits concentrically at the parent’s centroid; in Hex9 the parent centroid falls on the shared long edge of its two half-hexagons, so there is no central child and the 9 children form the shifted half-hexagon arrangement of §8. The shift is not a cost. A centred aperture-9 grid must descend one triangulation level below its own cells to describe their boundaries; Hex9’s half-hexagon is its own primitive at every level, and point

³⁴Generated by `libhex9 tools/dggs_survey_hex9.py` ($N = 5000$ cells per system, seed `0xc0ffee`); <https://github.com/MrBenGriffin/libhex9>. Cell areas are WGS84 geodesic areas (geographiclib); H3, S2 and A5 are measured through their own reference libraries. The surveyed Hex9 polygon is the hexagonal cell (the `x_cell` of §10a), the same footprint class as an H3 hexagon, not the internal half-hexagon.

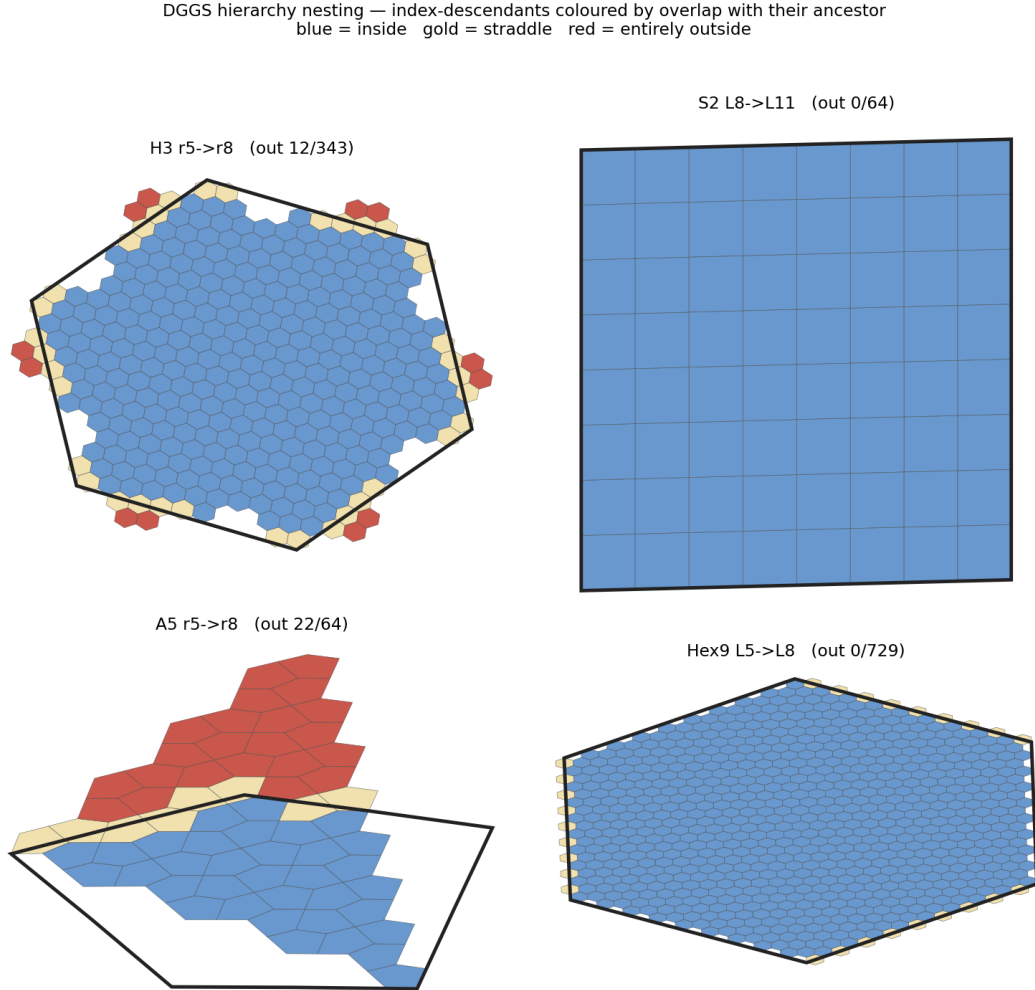


Figure 17: Hierarchy nesting across four DGGs. Each panel takes one anchor cell (heavy outline) and draws its index-descendants three resolutions finer — by that system’s *own* descendant relation — coloured by overlap with the anchor: blue lie inside, gold straddle the boundary, red lie **entirely outside** the ancestor whose address they extend. H3 (aperture-7, rotated each level) and A5 (equal-area, same-prefix children) *iterate* a single-level step and decohere: index-descendants drift out of the ancestor, A5 the more so. S2’s quadtree and Hex9’s per-layer contract do not iterate an offset — S2 nests exactly, and no Hex9 descendant lies outside the anchor (its gold cells are the genuine split hexes, each bound to its owner by the mode-0 convention). Generated by `docs/dggs/dggs_nesting.py` at anchor 40°N 3°W; the exact counts are anchor-dependent, the qualitative behaviour is not.

membership reduces to a fixed set of linear inequalities (§10a). Comparisons of aperture across systems should note this distinction.

We coin “shifted aperture 9” to be explicit about the offset. Each parent has exactly nine children — aperture 9 in the strict sense — but they carry a translational shift rather than sitting centred on the parent. This parallels H3, whose aperture-7 children carry a rotational offset between successive resolutions yet are still called aperture 7; we prefer to surface the offset in the name rather than leave it implicit.

Reference body. H3, S2, and HEALPix are defined on the sphere; ellipsoidal use requires an auxiliary-latitude transformation with its own distortion budget. Hex9 is defined against any two-axis reference ellipsoid directly: the warp is computed from geodesic areas on that body and recomputed for any other (§11d). Trained warps currently exist for WGS84 and the sphere; the C implementation (libhex9) is tuned for WGS84. WGS84 is the default, not a constraint.

Adjacency isotropy. As a graph, a hexagonal tiling gives every cell six neighbours related by symmetry, with no distinguished direction — where a quadrilateral grid distinguishes edge from corner neighbours (S2’s differ by a factor of $\sqrt{2}$), forcing distance and adjacency operations to treat the diagonal as a special case. This is a *combinatorial* property of the grid graph, not a metric one: once cells are placed on the ellipsoid, the base projection and the area-correcting warp make geodesic neighbour distances vary — **no DGGS, Hex9 included, is metrically isotropic on the globe**. What the hexagonal adjacency buys is algorithmic — traversal, k -ring, and neighbour queries carry no preferred axis. Hex9 and H3 share this adjacency isotropy; S2 does not (the diagonal $\sqrt{2}$); A5’s pentagonal cells additionally vary in shape (aspect ≈ 2.14).

Dual DGGS/CRS role. This is the row no other system marks “yes”, and it is the paper’s central claim (§9, §10e, Appendix A): a single Hex9 address is a DGGS zonal identifier when truncated at a level (OGC Topic 21), and in the limit a function recovers from it a point on the ellipsoid to arbitrary precision. We state this as the weaker, defensible claim — the addressing is *quasi-continuous* (§10e), not a strict ISO 19111 CRS: the recovery map is discrete-valued and jumps across a measure-zero set of seams, where continuity in the ISO sense fails. The distinction from the others is real but one of design, not exclusive capability: H3, S2, and HEALPix are bounded cell indices defined over a pre-existing CRS, and as used their identifiers label cells rather than recover positions. A cleanly-nesting index such as S2’s quadtree could in principle be extended to the same position-recovery role (§10e claims no uniqueness for the property). What is intrinsic to Hex9 is that a single address is, by construction, both the cell identifier and the position-recovery coordinate.

13. Implementation

Hex9 is implemented in **libhex9**, a C/C++ core library with several front-ends: a PostGIS extension, a Python accelerator binding, and a set of Python command-line tools. The earlier Python package (**hhg9**) remains as a readable reference implementation and produced the warp characterisation of §11b. Code artefacts use the prefix **h9** (e.g. **h9_encode**); prose uses Hex9 throughout.

13a. The partition cycle

The algorithmic heart of encoding is a repeated three-step cycle on octant coordinates: classify the point into one of the 9 child regions (the linear inequality evaluation of §10a); subtract the child’s origin; scale by 3. Each iteration emits one address digit and returns the coordinates to the original numerical scale, so there is no accumulating floating-point drift; the cycle is $O(L)$ per address, numerically stable, and exact in integer arithmetic. It is also the CRS limit of §10e

made operational: the offset removal composed with the $\times 3$ scaling is the inverse of a contraction mapping, and iterating it indefinitely converges to the encoded point.

Decoding runs the cycle in reverse, with one structural caveat: the split cells of §10b make the `x_list` right-to-left, so the decoder first reads the tail to fix the terminal mode and then resolves canonical parentage upward in a single pass.

13b. Address encoding

A Hex9 address packs into a single standard 128-bit UUID as 32 nibbles: 31 carry the hierarchy path (a root digit, then one base-9 digit per level, reaching layer 30) and the final nibble carries the tail (§10b). Because the canonical mode-0 fold (§10b) gives every cell a single representative, the UUID is **self-inverting**: a full address and any coarser **bin** of it are the same kind of object — each decodes, with no companion field, to its *own* cell’s representative point on the ellipsoid. The two tail forms (a marker nibble distinguishes a bin key from a full address) differ only in *which* cell is recovered: a bin inverts to the bin cell’s position, a full address to the terminal cell’s; neither recovers the original source point, which binning necessarily discards. (This supersedes an earlier design in which only the full address was reversible and a bin was an opaque, non-invertible key.)

The format drops directly into any database, PostGIS column, or API that accepts a standard UUID; bin keys at a common layer compare by plain equality or prefix, with no decoding step, and coarser bins derive from a full-depth address by truncation with tail canonicalisation (§10b, §12) rather than a raw digit cut — the distinction the split lineages require. The hierarchy reaches layer 30 — far below any geodetic resolution (a layer-30 cell is roughly 40 nanometres across) — so 128 bits is effectively unlimited precision in practice. The same canonical fit packs an address into a standard **64-bit** integer just as readily — as the Python library does — reaching about layer 14: cells about 1.7 m across, sub-metre positional quantisation (centimetre-scale at layer 17 with denser packing). The 128-bit UUID remains the default rather than a liability: PostgreSQL and comparable databases treat a UUID as a first-class, natively indexed type, so its width is no obstacle; the 64-bit form is offered for pipelines that require an integer column, not because the narrower key is intrinsically preferable.

13c. The core library and tooling

The `libhex9` C/C++ core implements the full pipeline: the AK base projection with its analytic Jacobian, the authalic warp (Clough–Tocher interpolant, Newton–Raphson inverse), encoding and decoding, canonical cell ancestry (§10b), k-ring neighbour computation, and cell-polygon generation, with the trained WGS84 warp embedded in the library. A Python accelerator (`hex9_ext`) exposes the core to Python at native speed with OpenMP batch encoding, and two Python command-line tools wrap the common workflows: `h9_csv` appends a reversible `h9_uuid` column — and, optionally, canonical bin or label columns — to a latitude/longitude CSV, streaming so it handles large files; `h9_choropleth` turns a point CSV into an adaptive Hex9 choropleth as a GeoJSON feature collection, with no PostGIS or QGIS in the loop. The `hhg9` Python package remains the readable reference: it produced the warp characterisation of §11b and the figures in this paper, with all grid operations vectorised over NumPy arrays. The canonical cell-ancestry operations of §10b — `h9_cell_parent` (one-level canonical parent, mode-0 convention) and `h9_cell_ancestor` (its composition) — are implemented in both libraries, distinct from the point-binning family, byte-identical across the pair for every cell through level 4, and verified by exhaustive enumeration: exactly nine canonical children per parent, for every cell, at every layer pair through level 5.³⁵

³⁵Verified by `experimental/cell_ancestor_verify.py`, together with the composition identity `ancestor(L-2) = parent-of-parent` and the §10b worked example (the canonical parent of cell 43585 is 4348, not its lineage cut

Beyond the repository’s own tooling, a Hex9 backend has been contributed to GeoPlegma,³⁶ the open-source project developing DGGS tooling and interfaces conformant to the OGC API — DGGS specification, whose DGGRS interface presents existing grid implementations (DGGRID, H3, S2) behind a single API. At the time of writing the integration is under review as a pull request.

13d. PostGIS extension

`libhex9` ships a PostGIS extension (`postgis_hex9`) that exposes the full surface in SQL: `h9_encode/h9_decode`, `h9_bin` and `h9_label`, the cell- and grid-polygon builders (`h9_cell`, `h9_grid`), neighbour queries (`h9_kring`, `h9_kdisk`, `h9_neighbors`), `h9_common_ancestor` for canonical roll-up, and `h9_adaptive` for population-driven mixed-resolution layers — so a Hex9 address is a first-class spatial key inside the database. It is built on the C/C++ core rather than Python, since most managed PostGIS environments treat Python as an untrusted language. (Distribution is handled directly through `libhex9`: PostGIS is winding down its third-party extension distribution programme.)

13e. On not registering a CRS

A PROJ/GDAL plugin was prototyped (`h9_boct`, a C++ port of the hierarchy walk and warp inversion), and the round trip matches the reference to within tens of nanometres. Two distinct objects could in principle be registered, and they have different standing.

The Hex9 *address system* is not registered, and should not be. The reason is the one developed in §10e: it is quasi-continuous, not a strict ISO 19111 CRS, so presenting it through the CRS machinery would overstate what it is. Its natural standards home is instead the DGGS track — OGC Topic 21 and its ISO publication as ISO 19170-1 — which exists precisely because discrete hierarchical grids do not fit the CRS mould. Applications use the core library and its bindings directly, treating Hex9 as the discrete addressing system it is.

The *AK+Warp projection* (§11) is a different matter. The `b_oct` space it realises is a continuous, real-valued coordinate space over a geodetic datum, with a smooth forward map and a convergent numerical inverse — the ordinary anatomy of an ISO 19111 projected CRS, untouched by the discreteness argument above. Its unusual features all have precedents in registered systems: projections without closed-form inverses invert iteratively throughout PROJ, interrupted layouts are established practice (Goode homolosine), and grid-backed coordinate operations are exactly what the NTV2 and geoid-model machinery distributes. Registering the AK+Warp projection as a projected CRS is deferred as future work (§15), not declined in principle.

13f. Inverse projection by guarded Gauss–Newton

The base-projection inverse — ellipsoid point to octant face coordinate — is the one numerically non-trivial step. `libhex9` provides a fast analytic-Jacobian solver for it, root-identical to the exact beam search in the smooth interior, at one analytic forward-and-Jacobian evaluation per iteration instead of several finite-difference evaluations each running an ECEF→geodetic (Bowring) step. The unknowns are the two face coordinates (f_x, f_y) ; the residual $r = P(f_x, f_y) - E$ is the 3-vector, in ECEF metres, between the AK forward map P and the target point E on the WGS84 ellipsoid. Because the forward map lands exactly on the ellipsoid, the least-squares root coincides with the true geodetic match — there is no auxiliary-latitude (Bowring) step inside the loop. Each iteration solves the 2×2 normal equations $(J^T J) \delta = -J^T r$ from the 3×2 Jacobian $J = \partial P / \partial (f_x, f_y)$, which is fully analytic: the chain of $\partial \text{AK} / \partial (u, v, w)$ (tangent-power derivatives and the ellipsoid-normalising quotient) composed with the affine $\partial (u, v, w) / \partial (f_x, f_y)$, and checked against central

4358).

³⁶<https://github.com/GeoPlegma/GeoPlegma>

differences. A backtracking line search (halving the step until $\|r\|^2$ decreases) makes it a strictly damped, guarded Gauss–Newton iteration; a seam/vertex guard and a failure veto route the thin skin around seams and vertices to the exact beam search, which remains the reference. The implementation is `libhex9/tools/newton_invert_aj.h`.

14. Applications

14a. Binning and density estimation

The primary use case is the one that motivated the quasi-authalic default: aggregating point observations into cells whose areas are comparable anywhere on Earth. Encoding is the $O(L)$ partition cycle; binning is truncation of the UUID with the canonical-ancestor derivation of §10b — a raw digit cut walks lineage, which at split lineages differs from ancestry, a discrepancy confined to the thinning band of §12. Because cell areas are uniform to within the warp residual, raw counts are densities up to a single global constant: no per-cell area correction, no latitude adjustment. Grids that are not equal-area (H3, S2) require explicit area normalisation for the same task, and the normalisation factor varies by location.

The same machinery serves raster products directly: a source callable samples each dataset at the cell centroid — one lookup per cell, apt at fine layers where a cell spans only a few source pixels. The figure below composites three such layers over a kilometre-scale window at the Emerald Bay trailhead, Lake Tahoe: NLCD 2024 land-cover classes fill the L12 cells (≈ 10 m), a DEM-derived hillshade modulates the relief through a fill two layers finer, and the L9–L11 cells are drawn as an overlay grid whose labels are the cells’ digits beneath the window’s shared prefix — the label elision of §10c working as a map graticule. The panel legend states what equal area guarantees: an L10 cell here is ≈ 1.22 ha at 68.5 m per side, and those numbers hold anywhere on Earth to within the warp residual.

Cells need not share a level. Because parentage is exact (§12), a single layer can mix resolutions — coarse cells where data is sparse, fine cells where it is dense — without T-junctions, seams, or interpolation between levels, and the mixed layer still rolls up losslessly. Refinement can therefore be driven by the data itself: subdivide a cell only while the population it contains stays within a target band, stopping when the count falls low enough or a depth limit is reached. The figure below shows this for Thimphu, Bhutan, where a population layer drives adaptive refinement across L5 to L12 — coarse L5 cells over empty terrain, refining to L12 along the inhabited valley floors. Each cell is shaded by population density (count per authalic cell area), so the fill is directly comparable across levels without area correction — a single coherent grid whose cells span seven levels at once. The layer is still a partition, not an overlay: each location lies in exactly one cell, because a finer cell is wholly contained in its coarser ancestor and the refinement keeps only the level its stopping rule selects — so cells of different levels tile without overlap.

14b. Spatial joins and multi-resolution analysis

Two datasets encoded to Hex9 addresses join on their canonical level-K bins: equality of bins is co-location at level K, computed without geometry. Bins derive from the stored full-depth addresses by the truncation-with-tail- canonicalisation of §12 — the route whose commutation with direct binning measures 0.00% over 10^4 points (L30 $\rightarrow$ L5, §12). Roll-up from L to any coarser K is canonical and lossless: every fine cell has exactly one ancestor at K, so aggregates recombine with

³⁸Generated by `examples/ex0252_geotiff.py` (Compositor/LayerSpec pipeline; each source sampled once per cell at its centroid). Sources: USGS Annual NLCD Land Cover 2024 (`Annual_NLCD_LndCov_2024_CU_C1V1`); hillshade computed from the CA Sierra Nevada DEM.

⁴⁰Rendered in QGIS via `h9_adaptive()` over the Bhutan population layer ([Meta Data for Good and Center for International Earth Science Information Network \(CIESIN\), Columbia University 2020](#)).

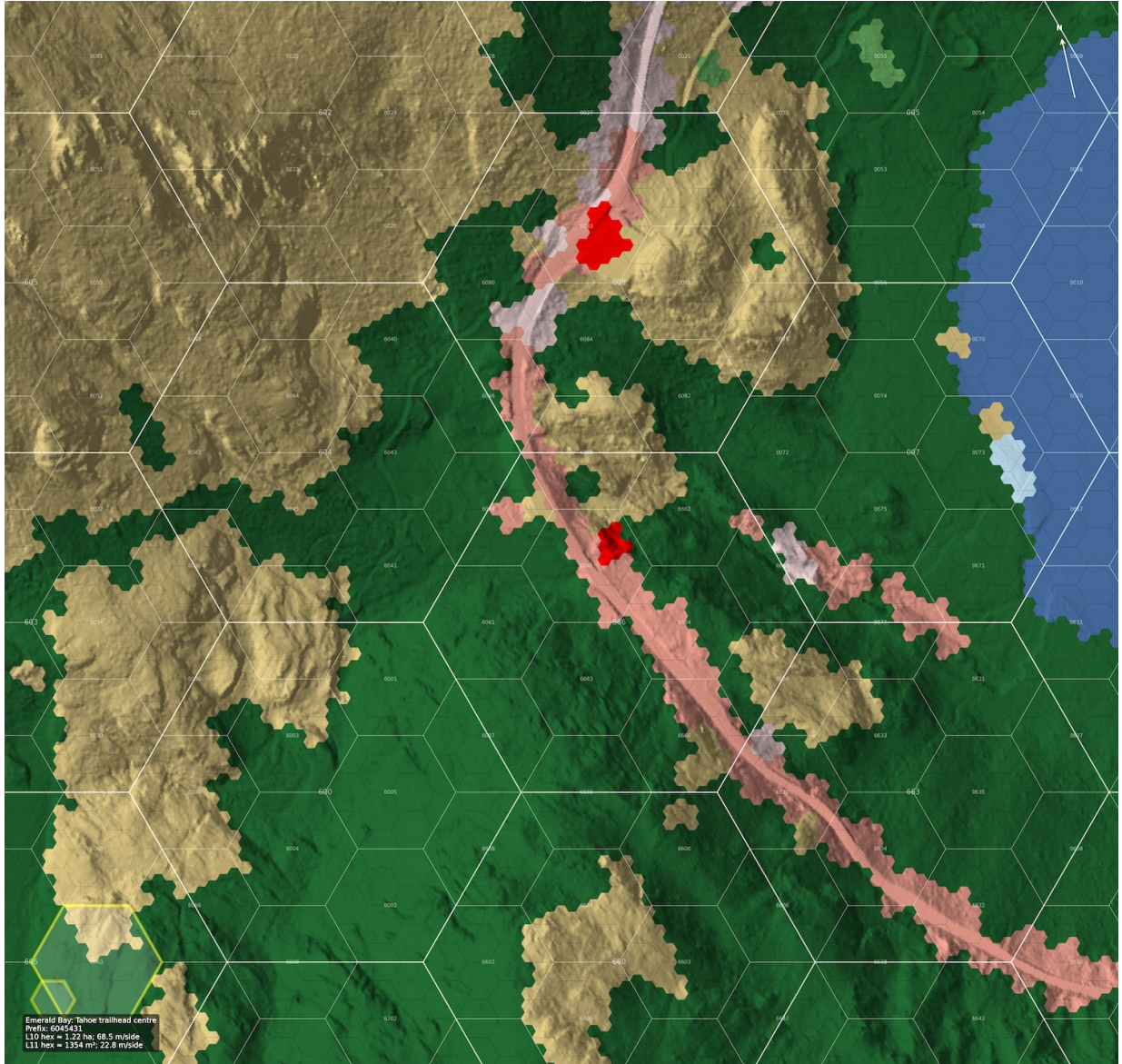


Figure 18: Land cover, relief, and addressing in one Hex9 composite (Emerald Bay trailhead, Lake Tahoe; kilometre-scale window). Three layers of one grid: USGS Annual NLCD 2024 land-cover classes fill the L12 cells; a hillshade derived from the Sierra Nevada DEM modulates at L14; the L9–L11 cells form the white overlay grid, labelled by their digit suffixes beneath the window’s shared prefix 6045431 (the label elision of §10c). The north arrow marks true north — the frame is the grid’s own net orientation, not north-up (§11c).³⁸

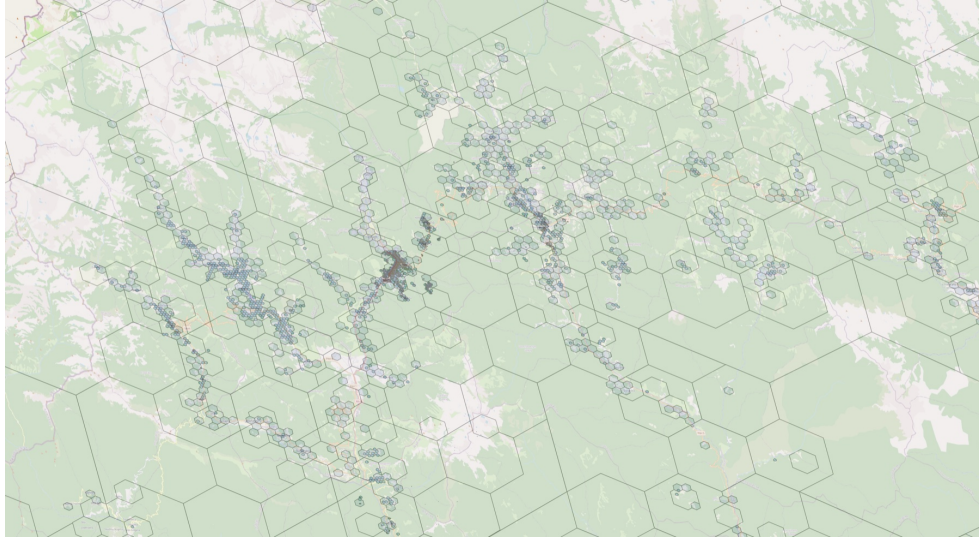


Figure 19: Thimphu, Bhutan — population-driven adaptive refinement spanning L5–L12 in one mixed-resolution Hex9 layer; fill is population density per authalic cell area, directly comparable across levels without area correction.⁴⁰

no double-counting and no residue. Its geometric fidelity is governed by the split-cell band of §12 — misassignment confined to adjacent cells, decaying threefold per level of separation, negligible for the deep-to-coarse roll-ups that dominate practice — where approximate-parentage systems misassign a constant fraction at any depth. The same property makes Hex9 addresses suitable as the spatial component of composite database keys, with standard B-tree indexes serving range and containment queries that would otherwise need a spatial index.

14c. Display and rendering

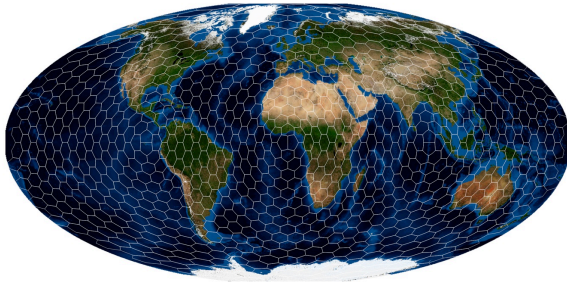
The canonical cell polygon is the regular hexagon in `b_oct`; everything else is reprojection (§11c). The equal-area realisation on the ellipsoid already departs from that regularity by the bounded amount of §12; a display in a conventional CRS adds more — cell edges are densified and reprojected, and the *additional* curvature that appears belongs to the target projection. Two practical notes. First, hexes straddling the antimeridian render incorrectly in naive geographic plots (drawn across the full map width); the defect is the renderer’s, not the data’s, and a bounds filter suffices. Second, at the six octahedral vertices the vertex cells are correct as constructed (§10f) and need no special-case rendering.

14d. Graticule alignment

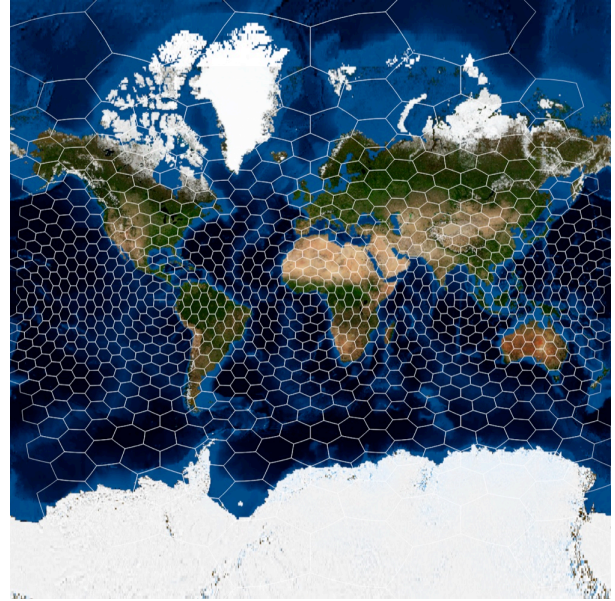
The octant face spans exactly 90° in latitude and longitude, and 360 carries two factors of 3, so trisection of the degree system is at least as natural as bisection — and more useful: bisecting 90° gives 45° , 22.5° , 11.25° (non-standard graticule values), while trisecting gives 30° and 10° , both standard. At 10° spacing the graticule fits the octant face in 9 clean intervals; L1 hexes span 30° , L2 hexes span 10° . In arc-minutes the octant ($5,400'$) trisects cleanly four times ($1800' \rightarrow 600' \rightarrow 200' \rightarrow 66.67'$) before breaking — deeper than bisection achieves. A 10° graticule is therefore the natural companion grid for Hex9 visualisations: the alignment is a consequence of the shared factor of 9 between base-360 and the ternary hierarchy, not a design choice in the projection.

14e. Other reference bodies

The addressing scheme, hierarchy, and interpolation machinery are ellipsoid-agnostic; only the Sinkhorn warp is computed against a specific body, at a one-time cost of roughly a day of unat-



(a) Mollweide (equal-area): areas preserved, shapes shear.



(b) Mercator (conformal): cells balloon toward the poles.

Figure 20: The same Hex9 layer reprojected — the distortion a conventional projection adds belongs to the map, not the cells (§14c). The native `b_oct` view, in which every cell is a congruent regular hexagon, opens the paper (Introduction); an equal-area projection (a) keeps cell areas comparable while shearing their shapes; a conformal projection (b) preserves local angles but inflates the polar cells without bound (the largest are the octahedral-vertex cells). Rendered in QGIS. Backdrop: NASA Blue Marble.

tended compute (§11d). WGS84 and GRS80 share a warp file in practice — their difference is negligible against the warp residual. Legacy ellipsoids (Bessel 1841, Clarke 1866) and planetary bodies with IAU reference ellipsoids (the Moon, Mars) require only their own warp files and their own prime-meridian conventions; the mathematical object is unchanged. Highly irregular bodies (Phobos, small asteroids) are out of scope, as for every current DGGS.

15. Conclusion and Future Work

The argument of this paper runs in one direction: from coherence requirements to structure, and from structure to location. Orientability selects the triangle; flat mode transport and vertex closure select even valence; regularity selects the octahedron; refinement commutativity selects odd apertures and the minimal aperture 9; the half-hexagon tiling enumeration — machine-verified end to end — leaves exactly one orientation per chirality. Setting the geometric realisation aside, precisely two genuine free choices remain in the entire construction: which chirality, and which bijection assigns the nine digits. Both are conventions in the same sense as the prime meridian: the grid does not depend on them, and a variant making the other choice is the same mathematical object.

Everything the system offers follows from that determinacy. Because no residual design freedom exists, cell identity is well-defined without reference to any prior coordinate system, and the address can serve as both a DGGS cell identifier and, in the limit, a position-recovery coordinate — quasi-continuous rather than a strict ISO 19111 CRS (§10e) — the dual role developed in §9 and §10e and formalised in Appendix A. The geometric realisation is a separable concern: the AK base projection and the optimal-transport warp place the abstract structure on WGS84 with

quasi-uniform areas, and either layer can be substituted without disturbing the grid.

Future work falls into four groups:

Warp refinement. A strictly authalic variant is achievable in principle ($\det J = 1$ everywhere) at the cost of unconstrained shear; the present quasi-authalic compromise is deliberate, but the trade-off curve — area residual against shape regularity, via the shape-dampening parameter — deserves systematic characterisation. The angular radius of the elevated-deviation zone around the six vertex singularities is a one-time, layer-independent measurement not yet made. Deeper-layer warps (L6+ derived rather than interpolated) would reduce the inter-mode residual further.

Implementation and maintenance. The core engineering work — the `libhex9` C/C++ library, its PostGIS extension, the Python accelerator, and the CLI tools (§13) — is in place; the forward effort is supporting, hardening, and improving it rather than building new infrastructure. Warp files for legacy ellipsoids (Bessel 1841, Clarke 1866) and at least one planetary body would demonstrate the generality claim concretely.

Verification. The R-rotation form of constraint B (§8) remains available as an independent cross-check of the tiling enumeration. The C1 continuity of the warp at octant boundaries is asserted from the construction and should be characterised explicitly.

Standardisation and community. Registration of the Hex9 address system as a PROJ/GDAL CRS was considered and is not being pursued: it is better identified as quasi-continuous rather than a strict ISO 19111 CRS (§10e, §13e), and the standards vehicle appropriate to its DGGS role is OGC Topic 21 / ISO 19170, not the CRS registry. The AK+Warp projection is the exception: as a continuous projected coordinate space it is an ordinary candidate for CRS registration (§13e), and that registration — together with a fuller cartographic treatment of the projection, including systematic distortion comparison against the Snyder and Lee octahedral projections — is planned as separate work. Discussions with PostGIS likewise suggested that it is better to pursue PostGIS integration as a separate third-party extension rather than look to integrate into the core; and this thought has been readily accepted. The path remains open engagement with the discrete-global-grid community around the existing `libhex9` implementation, rather than a formal standards submission. The OGC/ISO terminology mapping of Appendix B is offered in that spirit — as a bridge for readers coming from those standards, not a claim of conformance.

Appendix A — The CRS Limit

Claim. Every infinite Hex9 address sequence $\sigma = (a_0, a_1, a_2, \dots)$ uniquely determines a point on the WGS84 ellipsoid.

Setup. Let $\Omega \subset b_oct$ be the closed octant face: a compact convex triangle of diameter d_0 . One step of the Hex9 decode maps Ω into one of its 9 child cells by

$$p \mapsto \frac{1}{3}p + \text{offset}(C_i),$$

a Lipschitz contraction with ratio $1/3$. For an address prefix $(a_0, \dots, a_L)$, let $S_L \subset \Omega$ be the corresponding cell — compact and convex. Then

$$S_0 \supset S_1 \supset S_2 \supset \dots, \quad \text{diam}(S_L) \leq d_0 \left(\frac{1}{3}\right)^L \rightarrow 0.$$

Completeness. Each octant face is a closed bounded subset of $\mathbb{R}^2$ and hence a complete metric space; the S_L are closed subsets of it.

Cantor’s intersection theorem (nested compact sets with diameters tending to zero in a complete metric space) gives

$$\bigcap_{L=0}^{\infty} S_L = \{p^*\},$$

a single point of b_oct . Equivalently, via Cauchy sequences: choose any $p_L \in S_L$; for $L, M > N$ both p_L and p_M lie in S_N , so $d(p_L, p_M) \leq d_0 (1/3)^N \rightarrow 0$; the sequence is Cauchy, its limit p^* lies in every closed S_N , and uniqueness follows from $\text{diam}(S_L) \rightarrow 0$.

Lifting to the ellipsoid. The composition

$$b_oct \xrightarrow{\text{AuthalicWarp}^{-1}} b_raw \xrightarrow{\text{AK}} c_ell$$

is continuous: the warp is C^1 by the Clough-Tocher construction, and the AK projection is smooth on the octant interior and at its vertices (below). Continuous maps preserve limits, so p^* maps to a unique point on the ellipsoid.

Continuity at the octahedral vertices. The octahedral vertices are the points satisfying $|x| + |y| + |z| = 1$ and $x^2 + y^2 + z^2 = 1$ simultaneously — exactly the points with two zero coordinates. There the octahedron and the sphere coincide, the AK map reduces to the identity followed by axis scaling to the ellipsoid, and its apparent indeterminacy is a removable singularity; the Jacobian at these points is the well-defined limit of the linearisation. No actual singularity exists in the map.

Almost-everywhere bijectivity. The decode map $\sigma \mapsto p^*$ is surjective — every point is the limit of at least one address chain, namely its canonical encode — but not everywhere injective, and the failure has two distinct sources. The first is point-level and unavoidable: a point lying on a cell boundary at any level is the limit of more than one nested chain, one through each closed cell sharing that boundary — exactly as 0.5 admits both 0.5000... and 0.4999... in decimal. The set of such points is the union over all levels of the cell-edge sets: a countable union of measure-zero sets, hence measure zero on the ellipsoid. The second is cell-level and resolved by convention: 3 of the 9 child hexagons per level straddle two parents (§10b) and would otherwise admit two parent sequences for *every* point they contain — a positive-measure ambiguity if left unresolved. The canonical mode-0 convention selects one parent, making the canonical address of every cell unique and the encode map $p \mapsto \sigma$ single-valued everywhere. Away from the measure-zero boundary set, canonical infinite addresses and ellipsoidal points are in bijection.

The dual claim follows: σ truncated to L digits names a compact cell of area $510,065,622 \text{ km}^2 / (12 \cdot 9^L)$ on WGS84 (≈ 42.5 million km^2 at L_0 , 719.8 km^2 at L_5) — a DGGS zone in the sense of OGC Topic 21; σ in the limit names a point, a position recoverable by a function to arbitrary precision. This is the quasi-continuous coordinate role of §10e, not a continuous ISO 19111 coordinate system: the recovery map is well-defined and continuous in the decode direction (infinite address $\rightarrow$ point), but the address alphabet is discrete and the encode direction jumps across the measure-zero seam set just established. The contraction ratio $1/3$ is geometrically determined by the ternary subdivision, so the convergence rate is explicit: one address digit buys a factor-3 reduction in positional uncertainty.

Appendix B — OGC / ISO Terminology Mapping

This appendix maps Hex9 vocabulary to the two standards the dual claim of §9 engages: OGC Abstract Specification Topic 21 (20-040r3, 2021) ([Open Geospatial Consortium 2021](#)) — also

published as ISO 19170-1:2021 — for the DGGS role, and ISO 19111:2019 ([International Organization for Standardization 2019](#)) for the CRS role. The full taxonomy is maintained in the implementation glossary; the tables here cover the terms a standards reviewer needs.

Status codes: **C** — confirmed (the standard defines the term; the cited clause was checked against the published text); **A** — analogous (checked against the published text; the correspondence is real but approximate, as noted); — — Hex9-specific, with no standard equivalent. Clause numbers refer to OGC 20-040r3 and to OGC 18-005r4 (the OGC publication of ISO 19111:2019) respectively.

A vocabulary note on Topic 21: the 2021 edition does not simply rename *cell* to *zone* — it distinguishes them. A **zone** is a “particular region of space-time” (4.52), the unit that is identified; a **cell** is the “unit of geometry ... associated with a zone” (4.2). “While the terms cell and zone are often used interchangeably, strictly zone is the preferred term. Cell is entirely appropriate when specifically discussing a zone’s geometry or topology” (4.2, Note 3). This paper’s use of “cell” falls within that licence — Hex9 cells appear chiefly as geometry; where identification is meant, the Topic 21 term is *zonal identifier* (4.50).

Hex9 → OGC Topic 21 (DGGS)

Hex9 term	OGC Topic 21 term	Status
Hex9 (the system)	discrete global grid system (4.13)	C
x_grid at level L	discrete global grid — one tessellation per level (4.12)	C
x_cell	zone (4.52); its geometry, a cell (4.2)	C
x_adr / x_list	zonal identifier (4.50)	C
octant (8 seed faces)	face of the base unit polyhedron (4.27)	C
12 root x_cells	initial discrete global grid, refinement level 0 (4.27, 4.37)	C
refinement level L	refinement level (4.37)	C
aperture 9 (shifted)	refinement ratio 9 (4.38; Note 3 there records “aperture” as earlier DGGS-literature usage)	C
binning (h9_bin)	quantization (4.36)	C
parent/child x_cell	parent cell / child cell (4.33, 4.4)	C
split x_cell (digits 6–8)	child cell “overlapped by multiple parent cells” (4.4, Note 1)	C
x_dig	— (ordinal of a child within its parent; no Topic 21 term)	—
AK+Warp realisation	surface model of the Earth (4.27); cf. Part 1’s Equal Area Earth Reference System (Hex9 is quasi-equal-area, §11b)	A
mode (0/1 parity)	— (internal face orientation)	—
c2 edge label	— (sub-face structure)	—
d_cell (half-hexagon)	— (sub-cell primitive)	—

Hex9 → ISO 19111:2019 (referencing by coordinates)

Hex9 term	ISO 19111 term	Status
WGS84 reference ellipsoid	ellipsoid / geodetic reference frame (3.1.34)	C
Prime Meridian anchoring (Axiom 7)	prime meridian (3.1.50)	C

Hex9 term	ISO 19111 term	Status
<code>x_addr</code> at a fixed level	spatial reference “in the form of a label, code” (3.1.56); a geographic identifier in the ISO 19112 sense (via Topic 21 4.50, Note 1)	C
<code>x_addr</code> carried to the limit (§10e, App. A)	approaches the role of a coordinate — “one of a sequence of numbers designating the position of a point” (3.1.5) — quasi-continuous, not a strict ISO coordinate	A
Hex9 as a locating system	coordinate reference system (3.1.9) — <i>quasi</i> , see §10e	A
<code>b_oct</code> over WGS84, realised by AK+Warp	projected CRS (3.1.51, §9.2.2); map projection — “coordinate conversion from an ellipsoidal coordinate system to a plane” (3.1.40)	C
encode / decode (point ↔ address)	coordinate conversion (3.1.6) for the geodetic ↔ <code>b_oct</code> leg; the address leg assigns a spatial reference (3.1.56), not a coordinate tuple	A
octant 2D plane coordinates	coordinate system / axes (3.1.11)	C

The dual role of §9, §10e, and Appendix A is the bridge between these two tables: one Hex9 address is a Topic 21 zonal identifier when truncated at a level and, in the limit, converges to naming a point — the role ISO 19111 reserves for coordinates (3.1.5). The standards themselves mark the boundary the dual claim crosses: Topic 21 notes that in a DGGS “locations have dimension greater than one, and so are not points” (4.31, Note 2), and that a zonal identifier “may be a geographic identifier” in the ISO 19112 sense (4.50, Note 1) — a label, not a coordinate. Hex9’s finite-level behaviour sits squarely in that zonal world; the limit behaviour of Appendix A is what reaches toward ISO 19111, and we make the weaker, defensible claim for it — *quasi-continuous*, by analogy with quasi-authalic — rather than asserting a strict ISO 19111 CRS: the address space is discrete and the point→address map is discontinuous on a measure-zero seam set (§10e), so continuity in the ISO sense does not hold.

Figures

Full figure specifications, production notes, and final captions are maintained in `paper_figures.md`; images live in `paper_figures/`. The index below lists the figures and their section anchors.

#	Section	Working title
F1	§8 / §2	The 49 tilings, Hex9 pair highlighted
F2	§8	The two equilateral tilings (L/R chiral pair)
F3	§10.0	Anatomy of one parent triangle: $t / c_2 / d / x$
F4	§10a	x-layer: <code>x_dig</code> 0–8 children, high-trit colouring + shared c-regions
F5	§10a	Classifier (c-layer): 96-slot <code>c_grid</code> , three band families → <code>c_dig</code>
F6	§10b	London / Prime Meridian split-cell example

#	Section	Working title
F7	§10d / §10f	Sibling adjacency: internal vs cross-parent edges
F11	§11	Pipeline diagram $b_raw \rightarrow \text{warp} \rightarrow b_oct \rightarrow \text{address}$
F12	§11b	Area-deviation globe, $\pm 1\%$ capped (magnitude view)
F12b	§11b	Area-deviation Mollweide, clipped scale (pattern view: seam skeleton)
F12c	§11b	Vertex census — the three vertex classes, one panel each (ex0124)
F13	§11c	Tissot indicatrices on the b_oct butterfly net (warp applied)
F15	§14a	Land cover + hillshade + address-grid composite (Emerald Bay, ex0252)
F16	§14c	Same layer in b_oct / Mollweide / Mercator
F20	§2 / §7 / §8	Wallpaper group of the d_cell tiling (p31m)
F21	§9 / §10 (graphical abstract)	Seed solid (24 d_cell facets) + 12 root x_cells
F23	§14a	Thimphu adaptive refinement (L5–L12), population density choropleth

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